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Mountz

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(54) **SEAT SUPPORT FOR CHILD SWING**

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(73) Assignee: **Wonderland Switzerland AG**, Steinhausen (CH)

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(22) Filed: **Jul. 20, 2023**

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Related U.S. Application Data

(60) Provisional application No. 63/414,217, filed on Oct. 7, 2022, provisional application No. 63/401,091, filed on Aug. 25, 2022, provisional application No. 63/369,802, filed on Jul. 29, 2022.

(51) **Int. Cl.**
A47D 13/10 (2006.01)

(52) **U.S. Cl.**
CPC *A47D 13/107* (2013.01)

(58) **Field of Classification Search**
CPC A47D 13/10; A47D 13/107
USPC 297/DIG. 11
See application file for complete search history.

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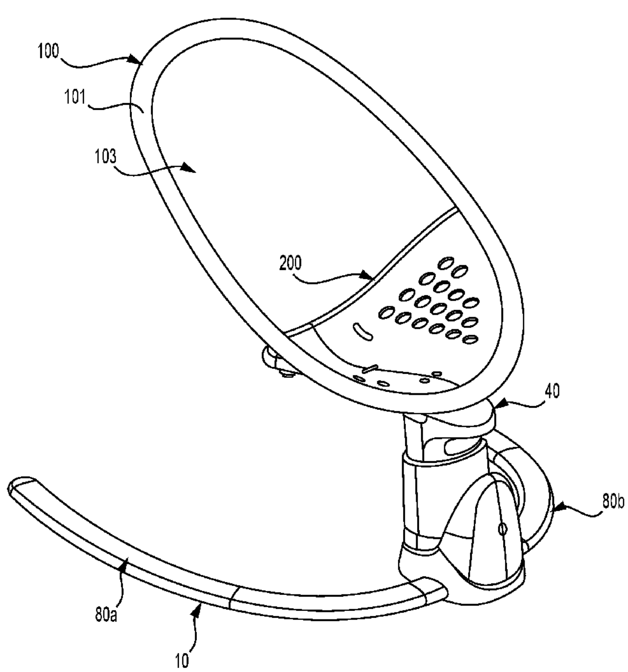
Primary Examiner — Milton Nelson, Jr.
(74) *Attorney, Agent, or Firm* — Fay Kaplun & Marcin, LLP

(57) **ABSTRACT**

A child swing has a base, a column, and a seat. The seat comprises a seat rim and a seat support. The seat rim is connectable to the column to support the seat above a floor. The seat rim defines a seat opening. The seat support comprises a rigid material and is attached to the seat rim such that movement between the seat support and the seat rim is substantially prevented. The seat support defines a recess on one side of the seat opening. The seat support is positioned relative to the seat rim such that when a child is introduced through the seat opening from an opposing side of the seat opening, the seat support supports the child within the recess.

21 Claims, 34 Drawing Sheets



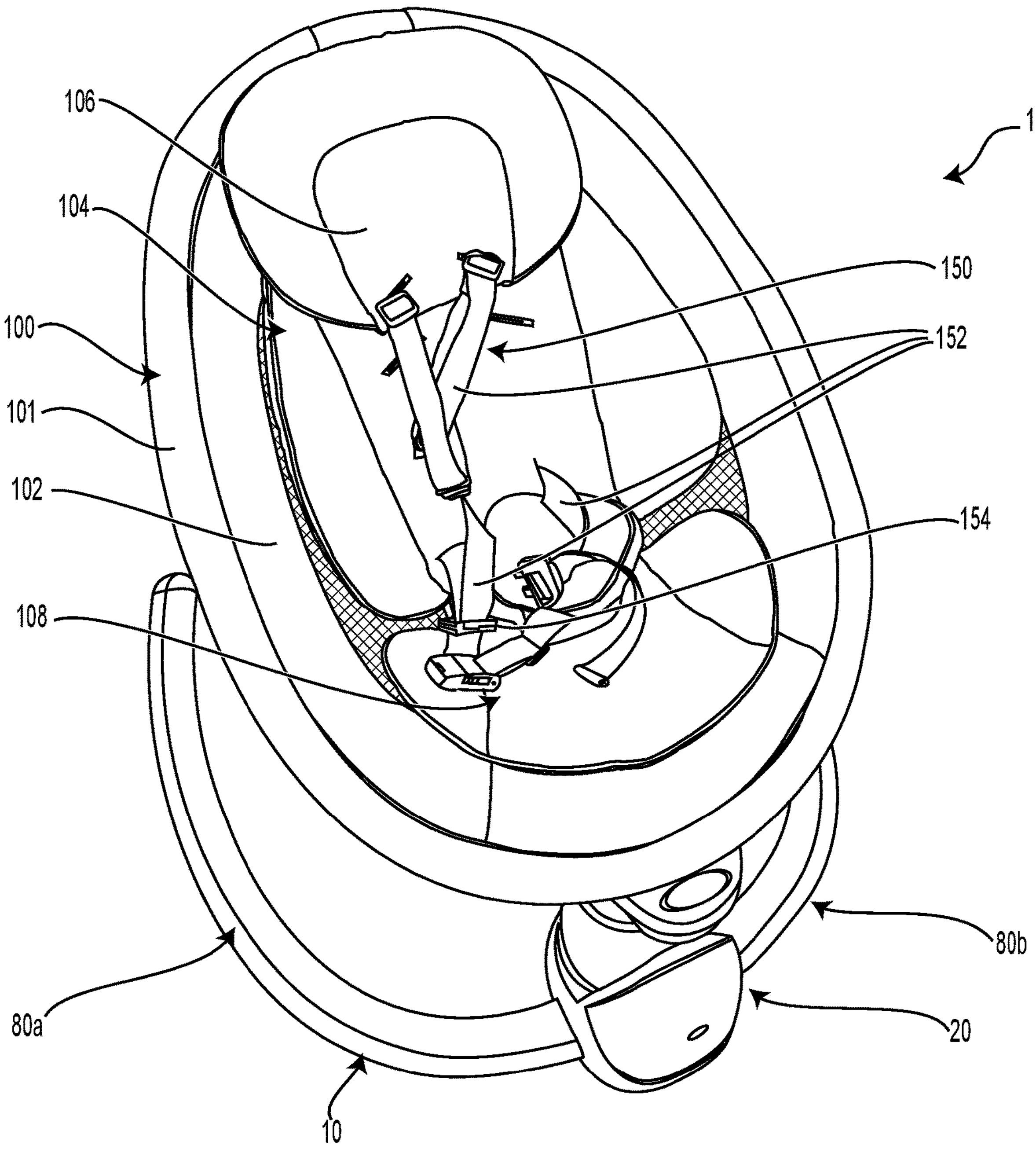


FIG. 1

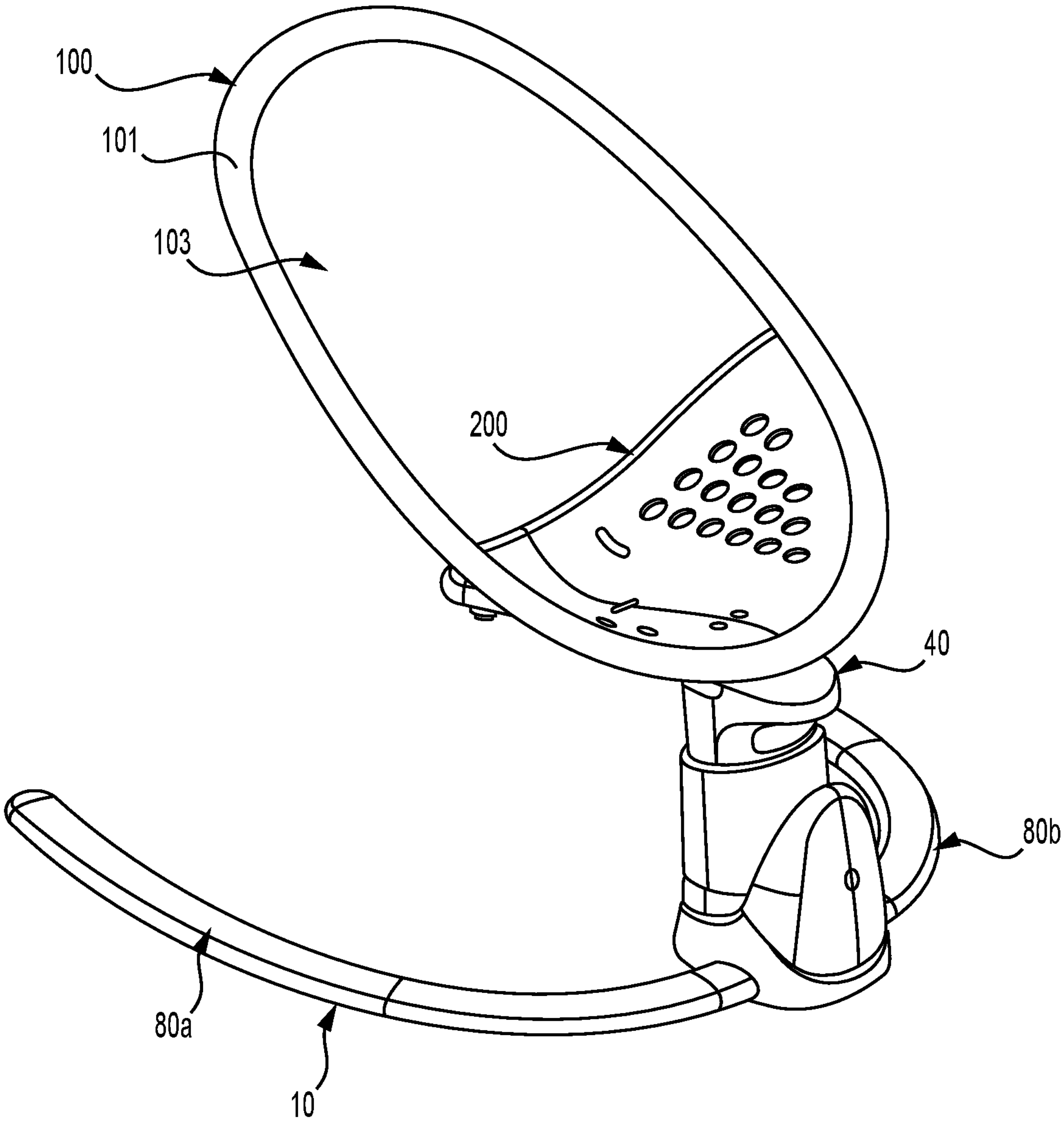


FIG. 2

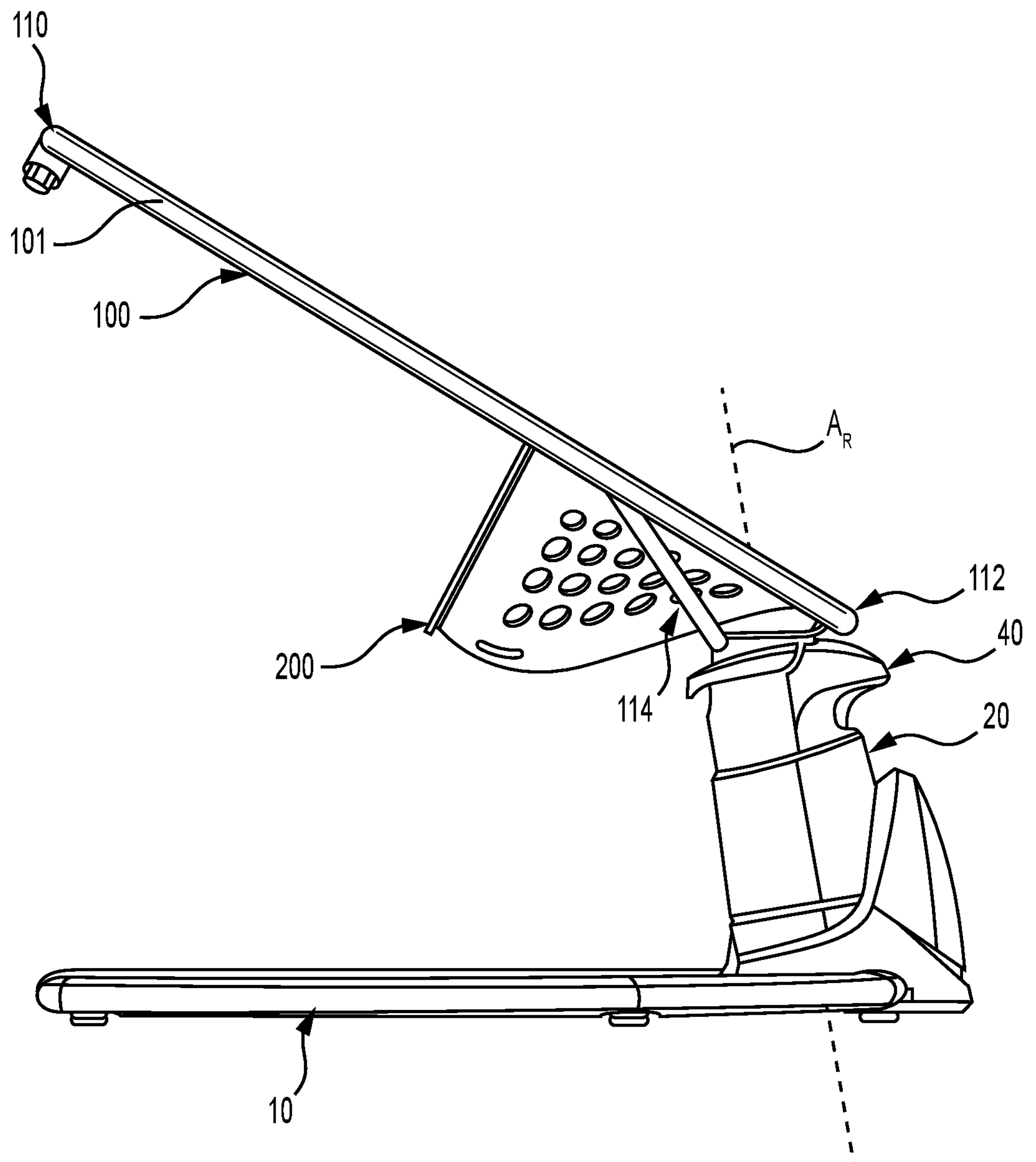


FIG. 3

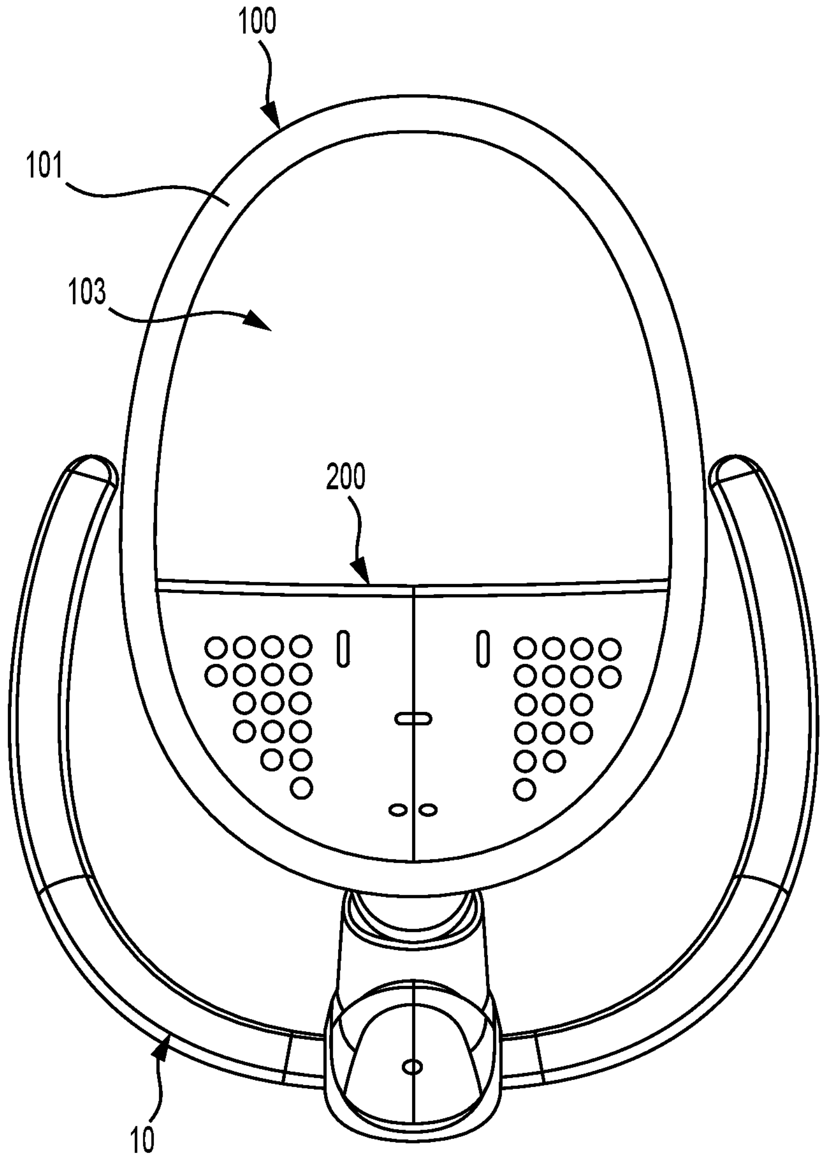
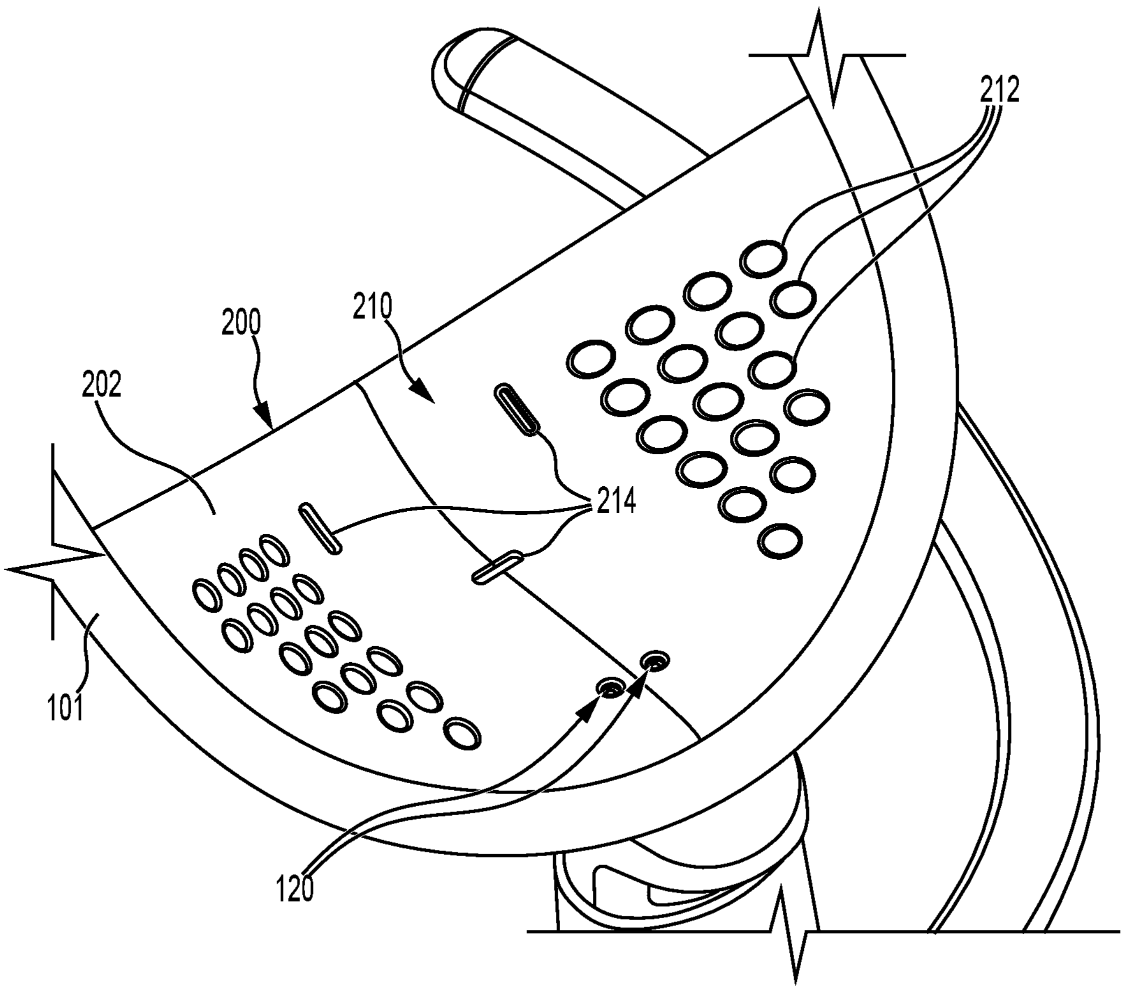


FIG. 4



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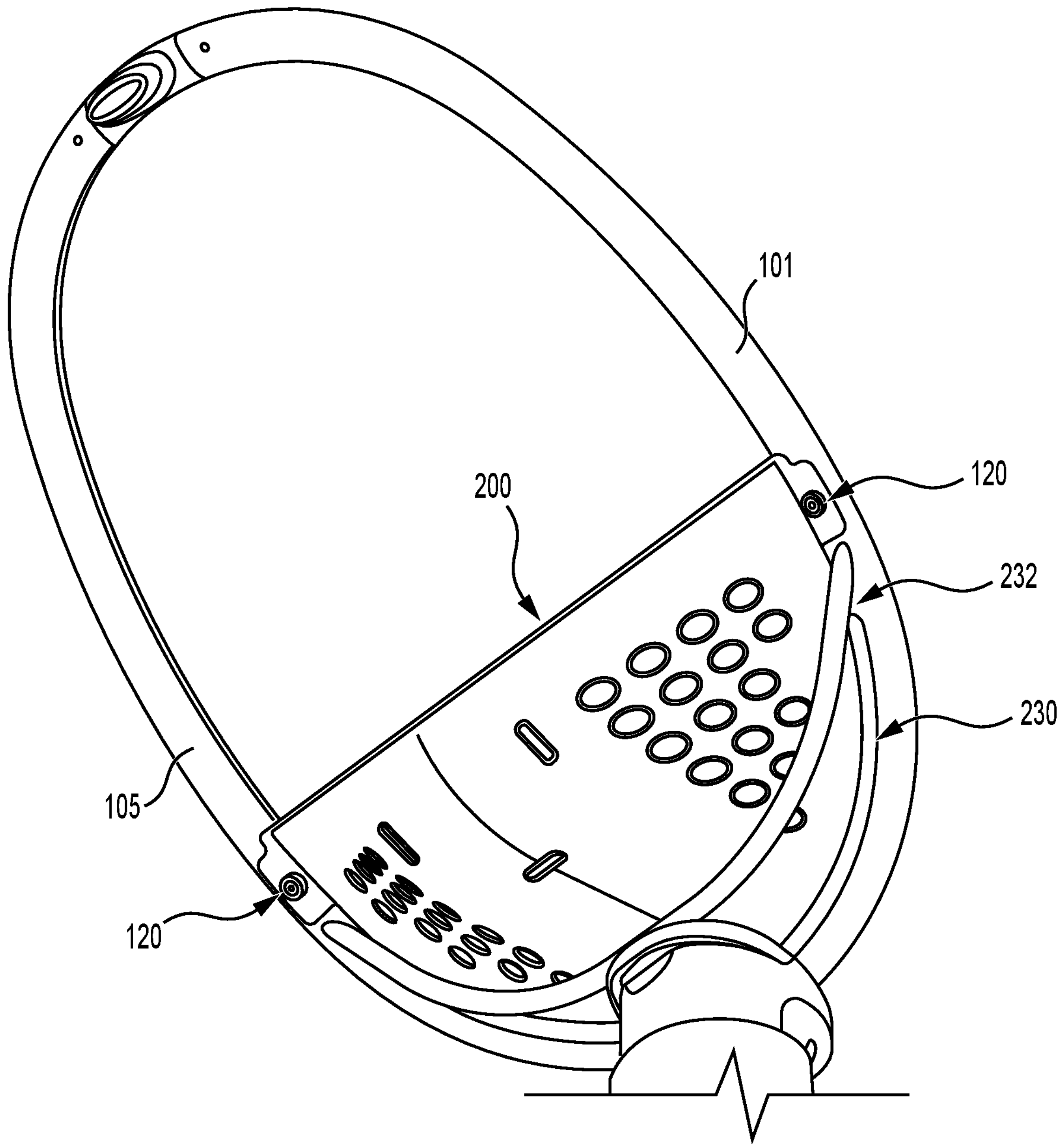
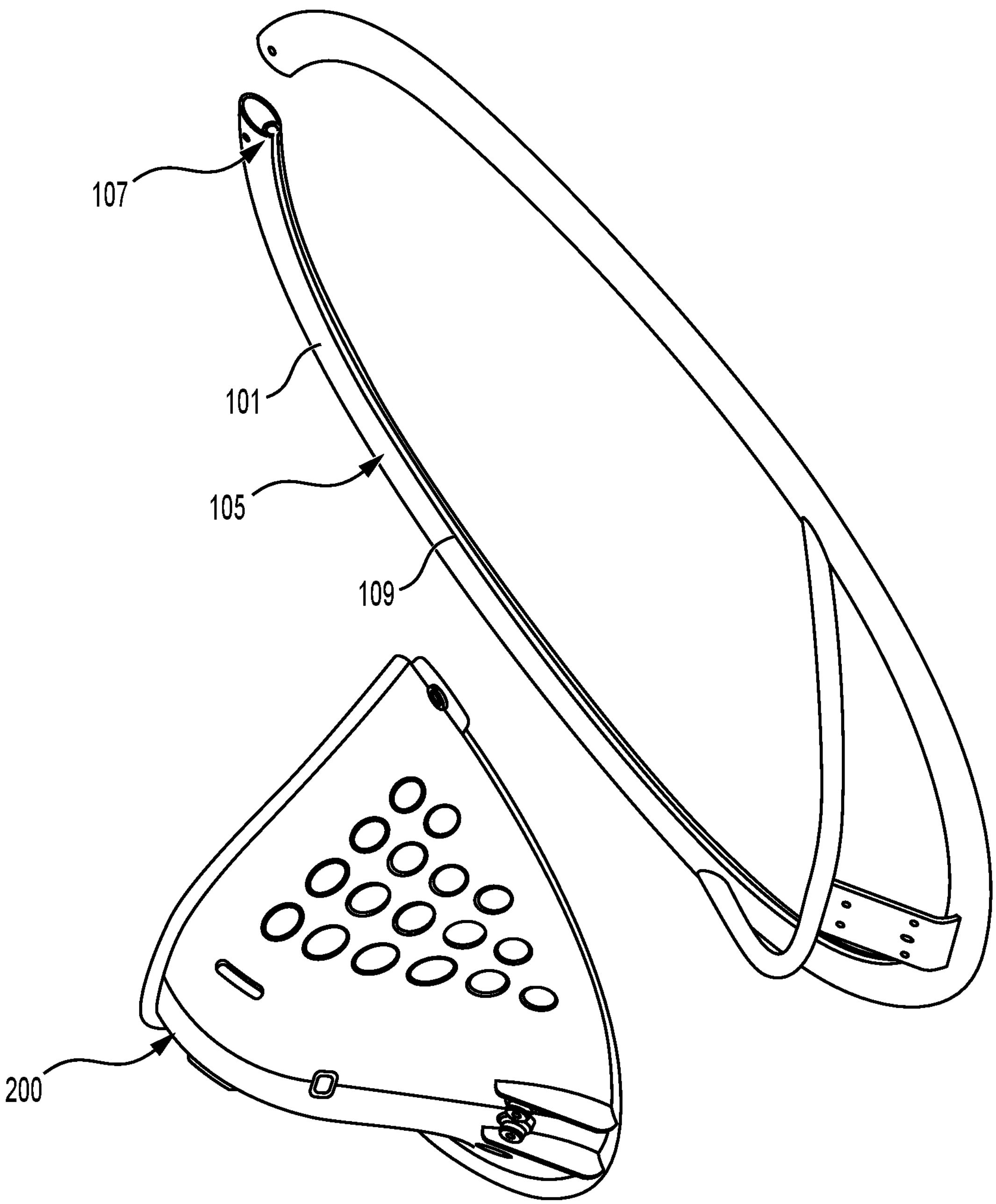


FIG. 6



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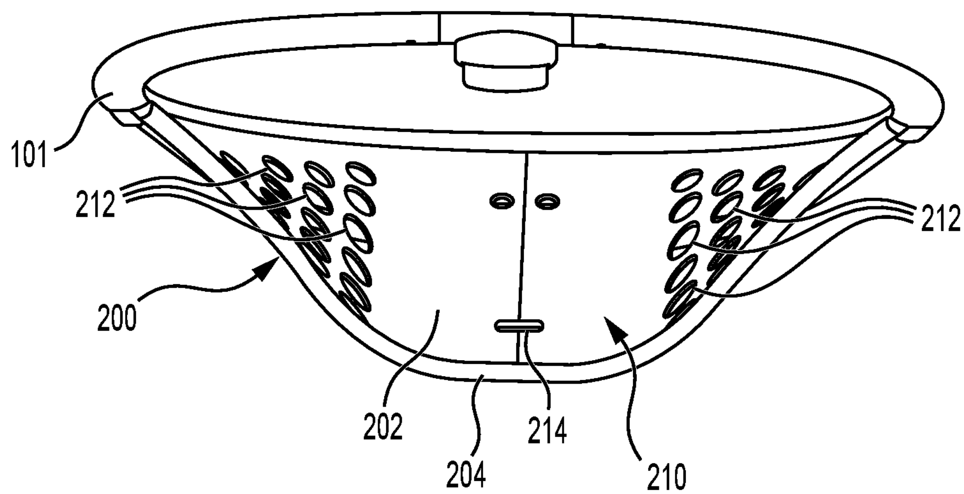

FIG. 8

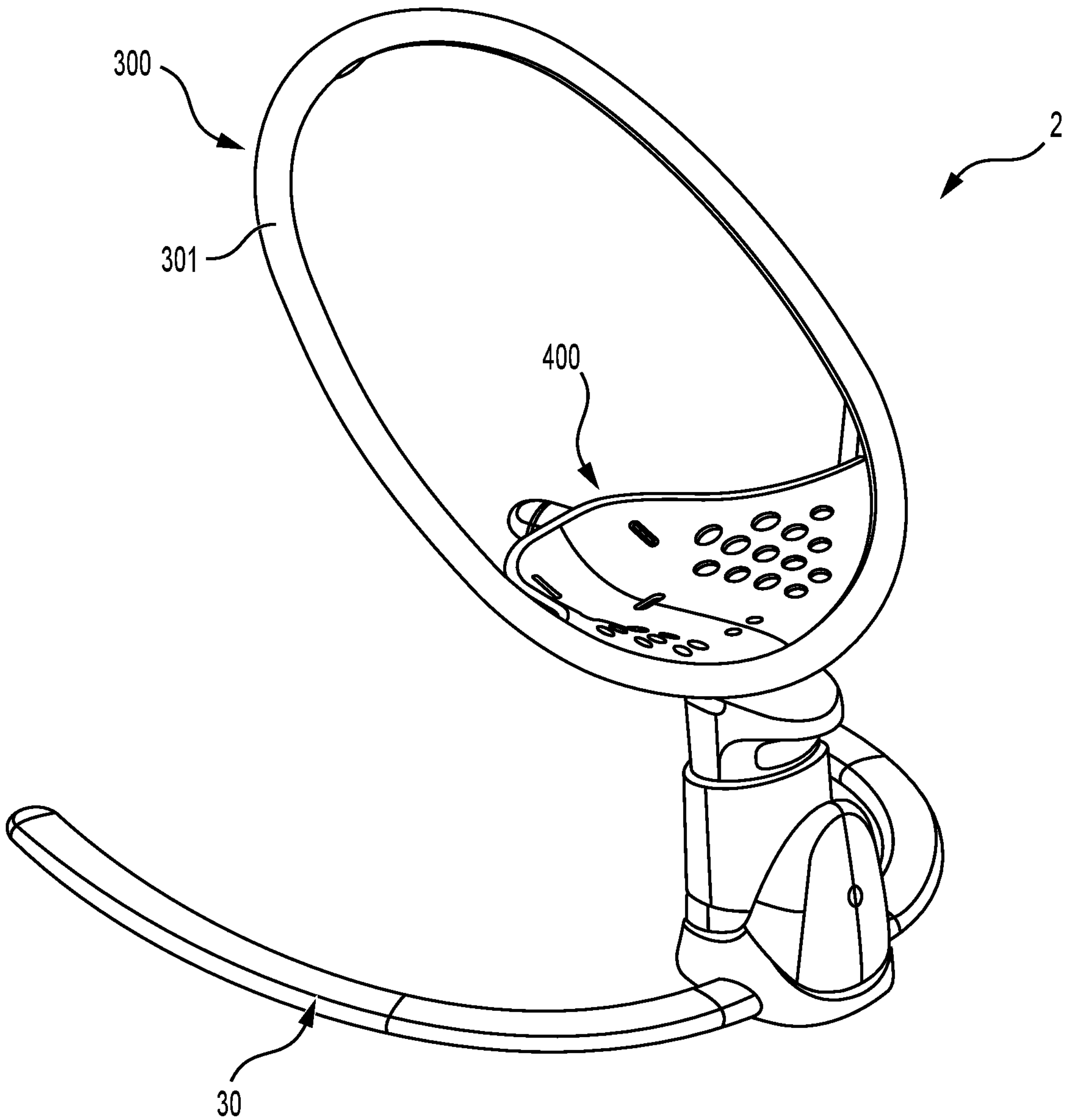

FIG. 9

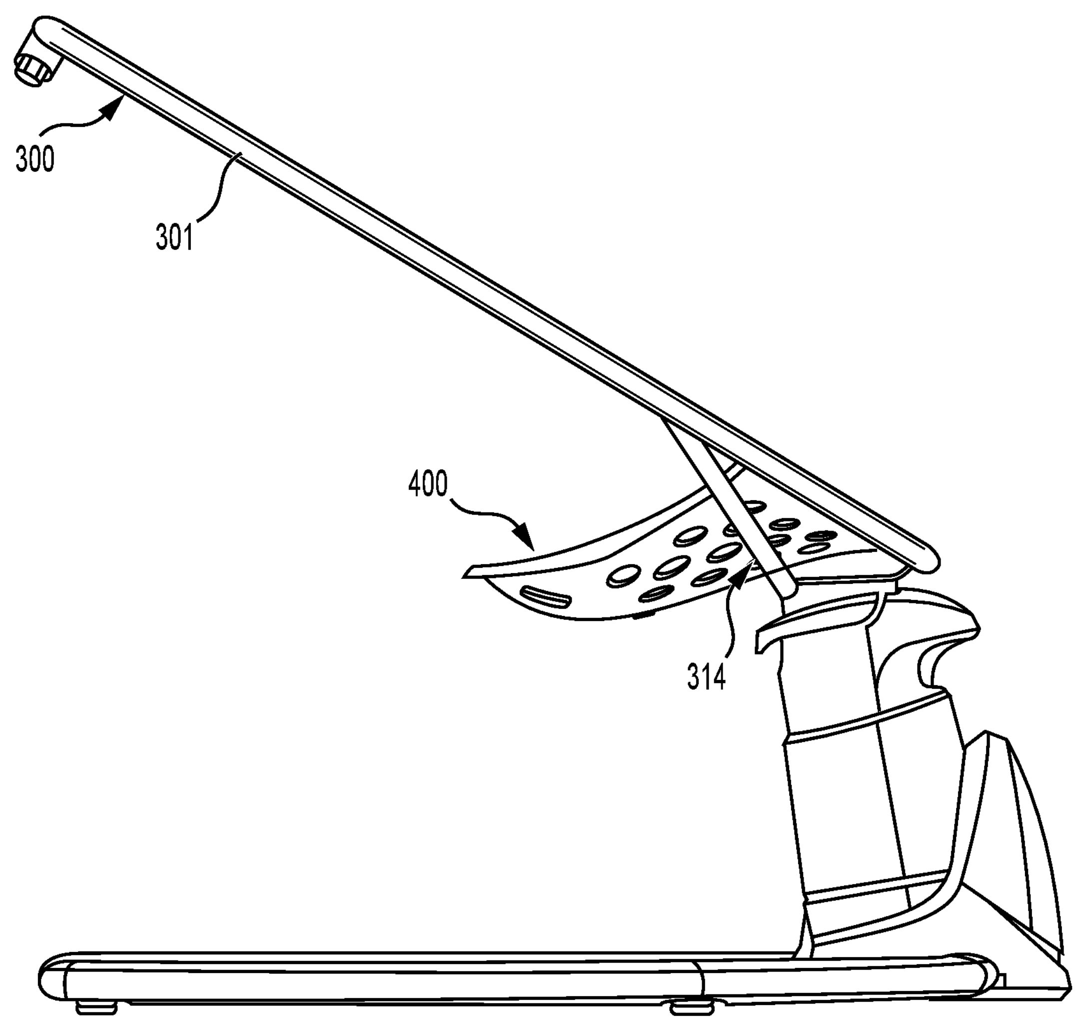

FIG. 10

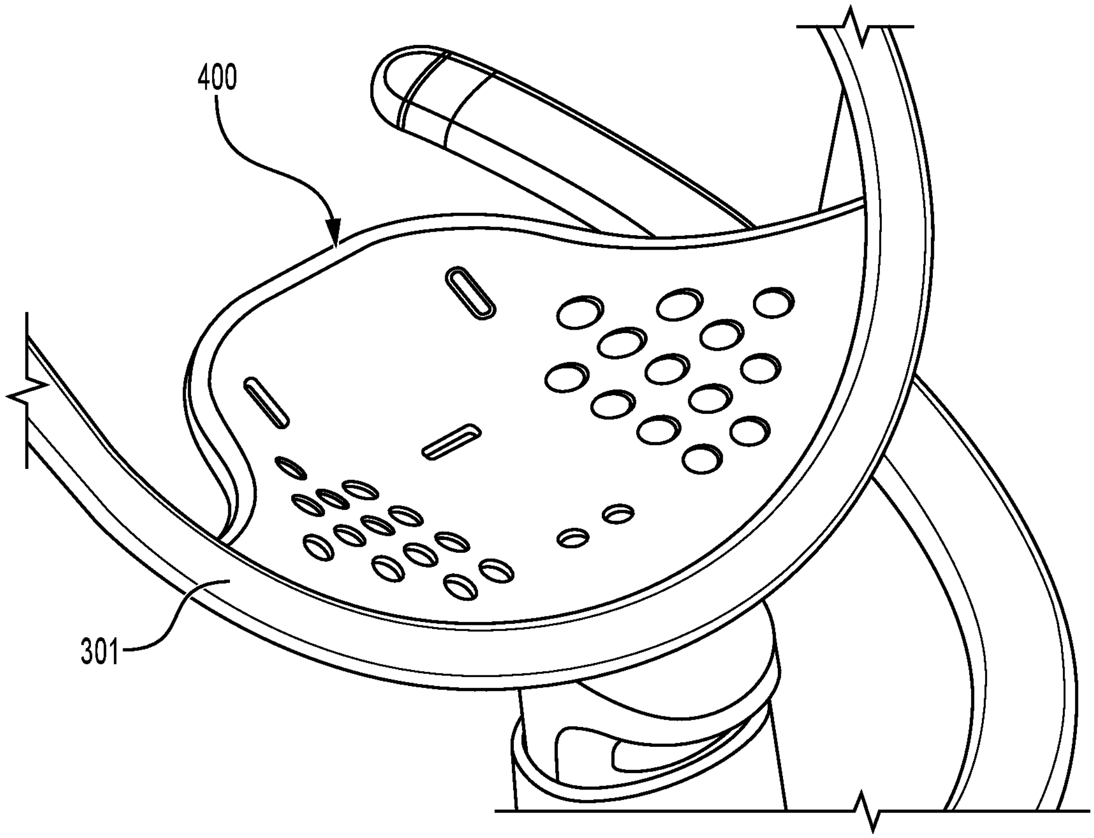

FIG. 11

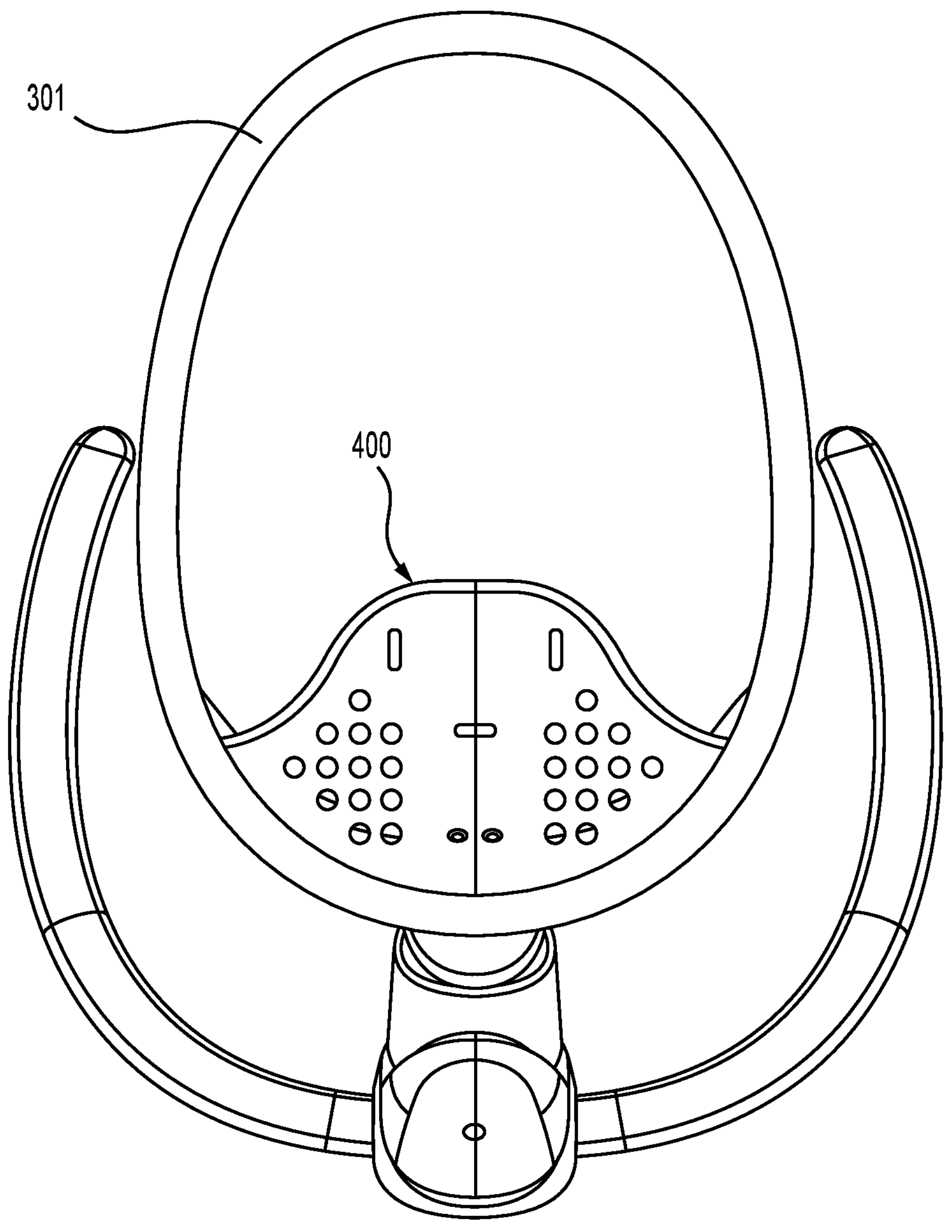

FIG. 12

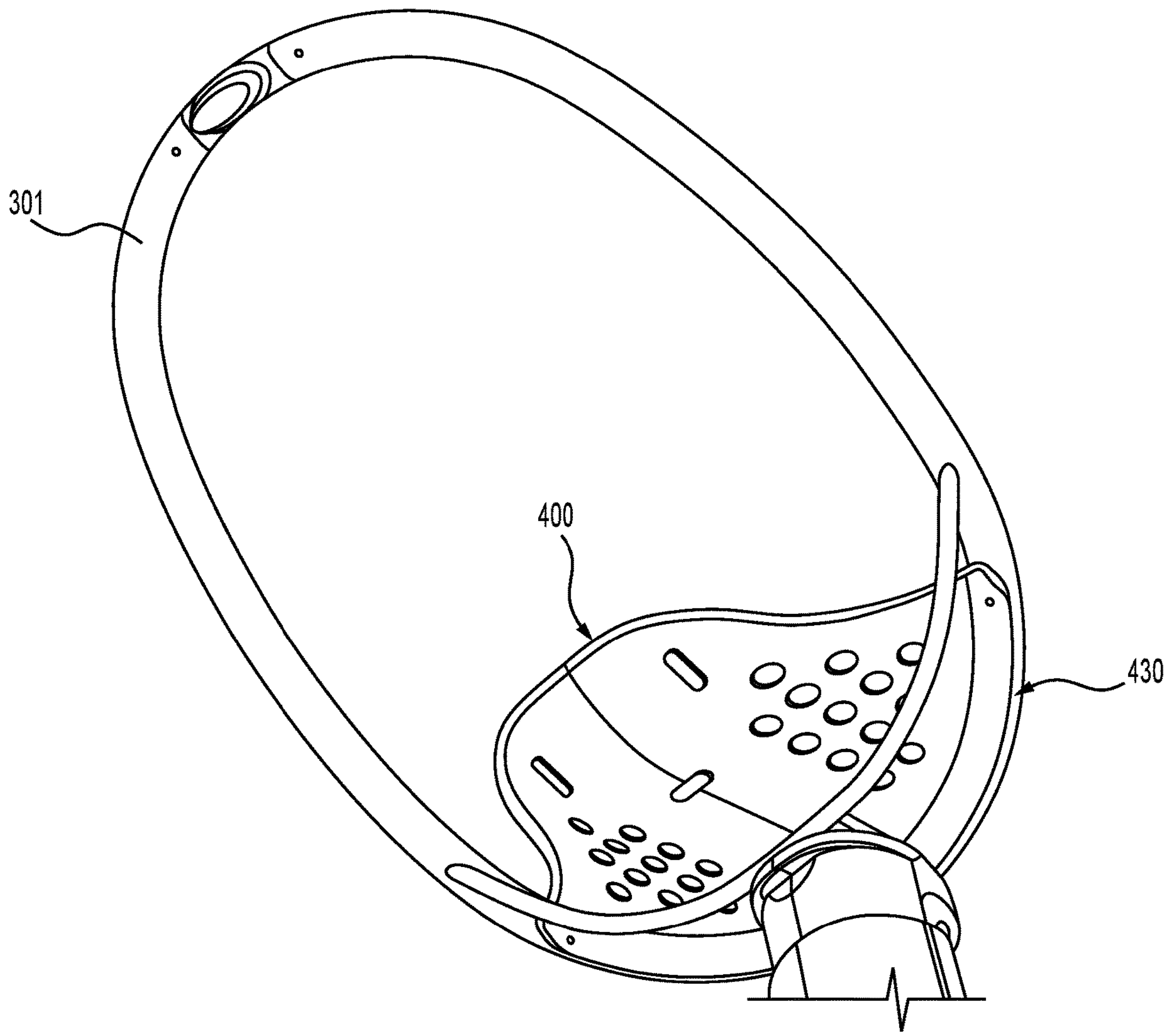

FIG. 13

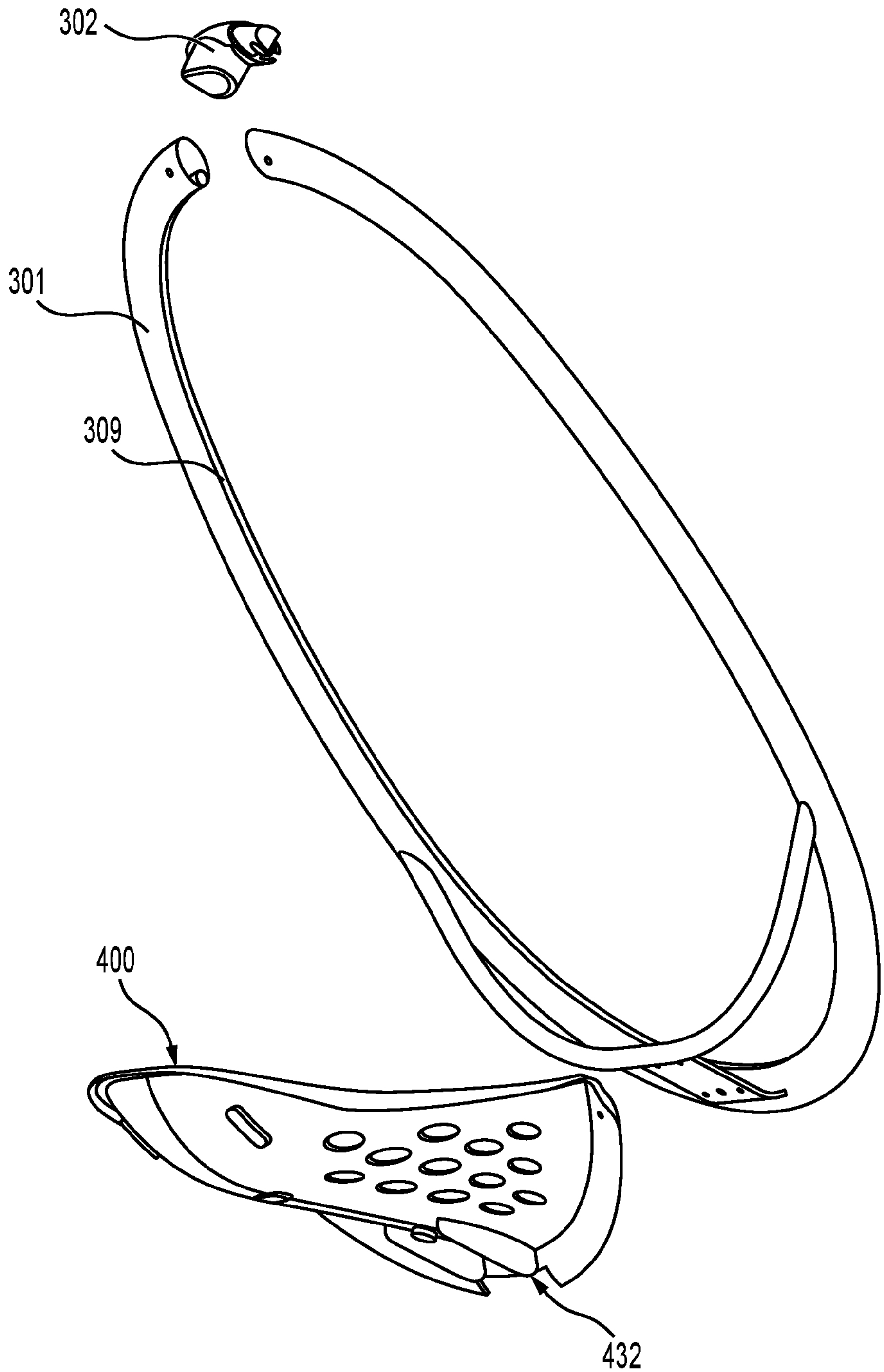

FIG. 14A

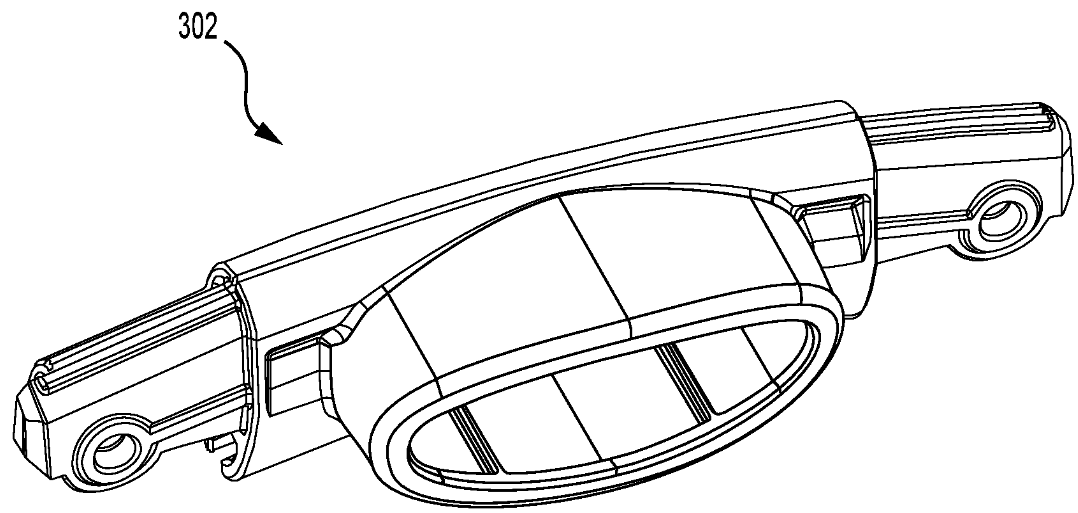

FIG. 14B

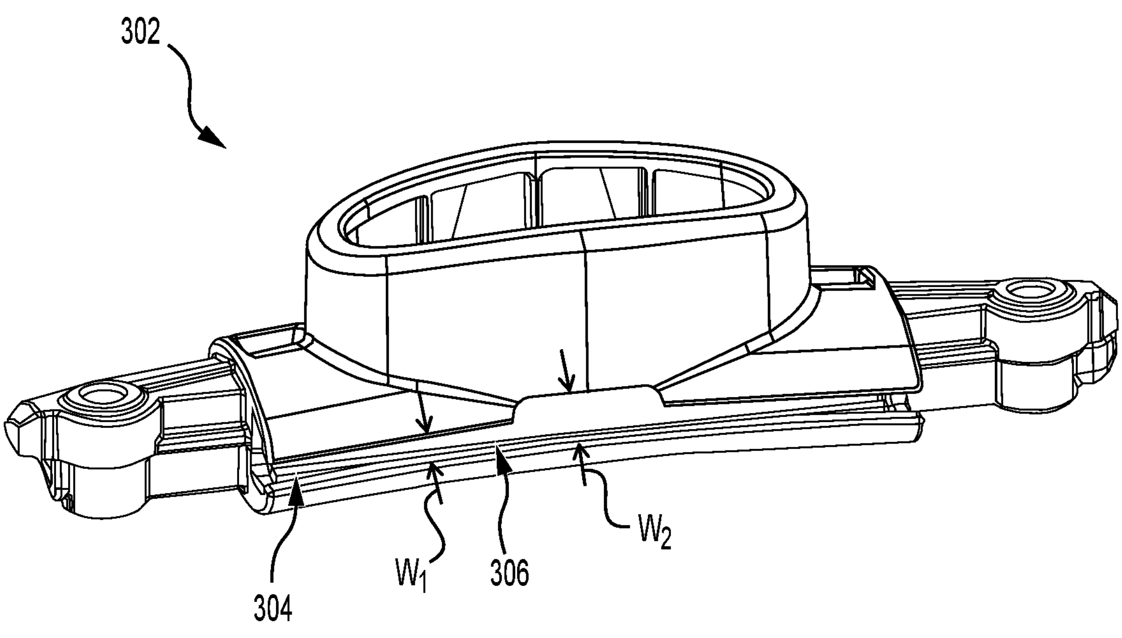

FIG. 14C

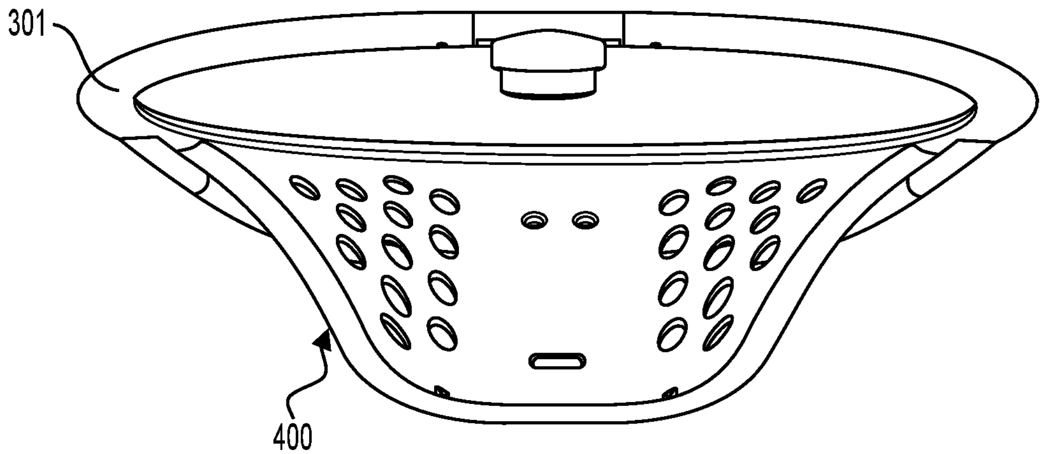

FIG. 15

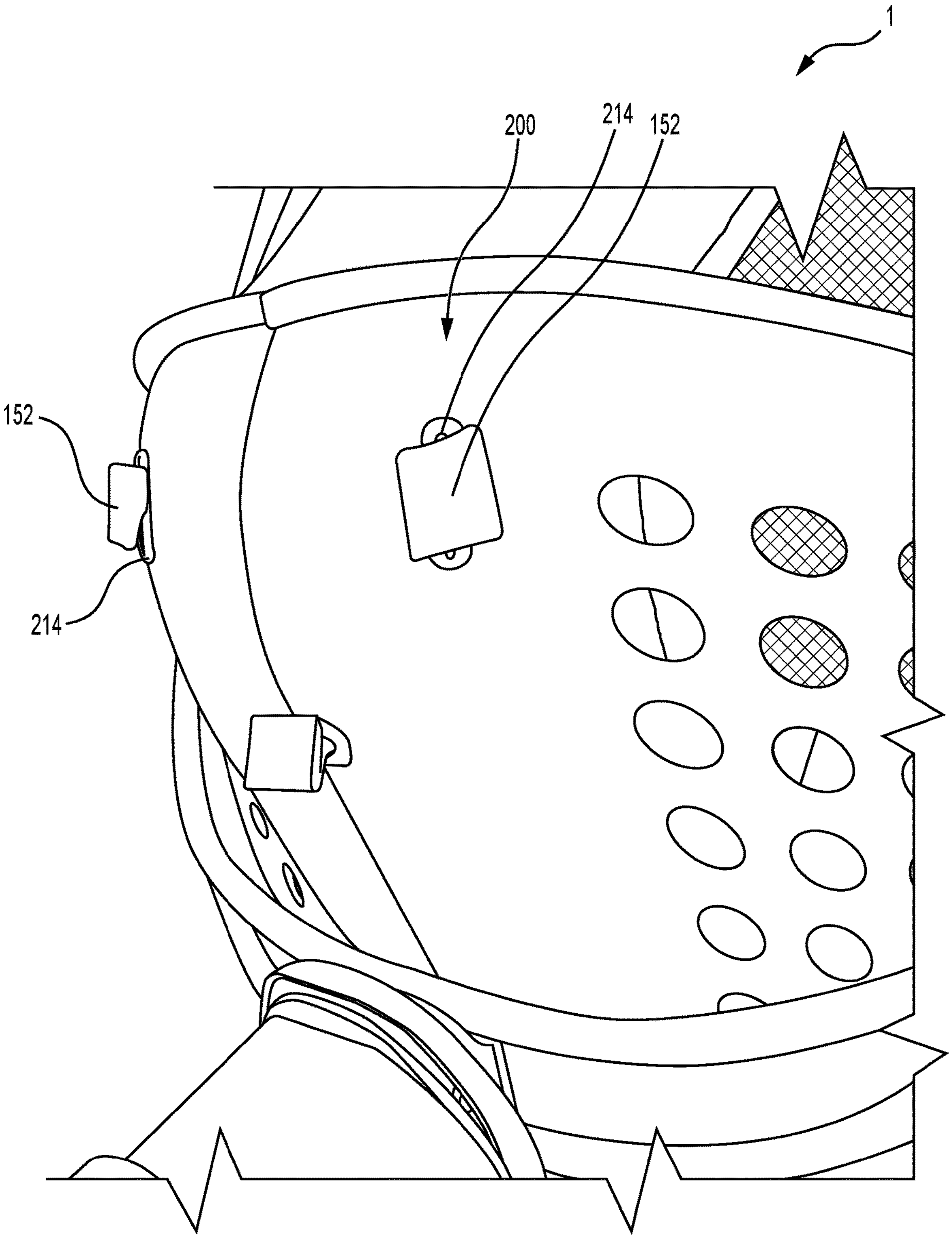

FIG. 16

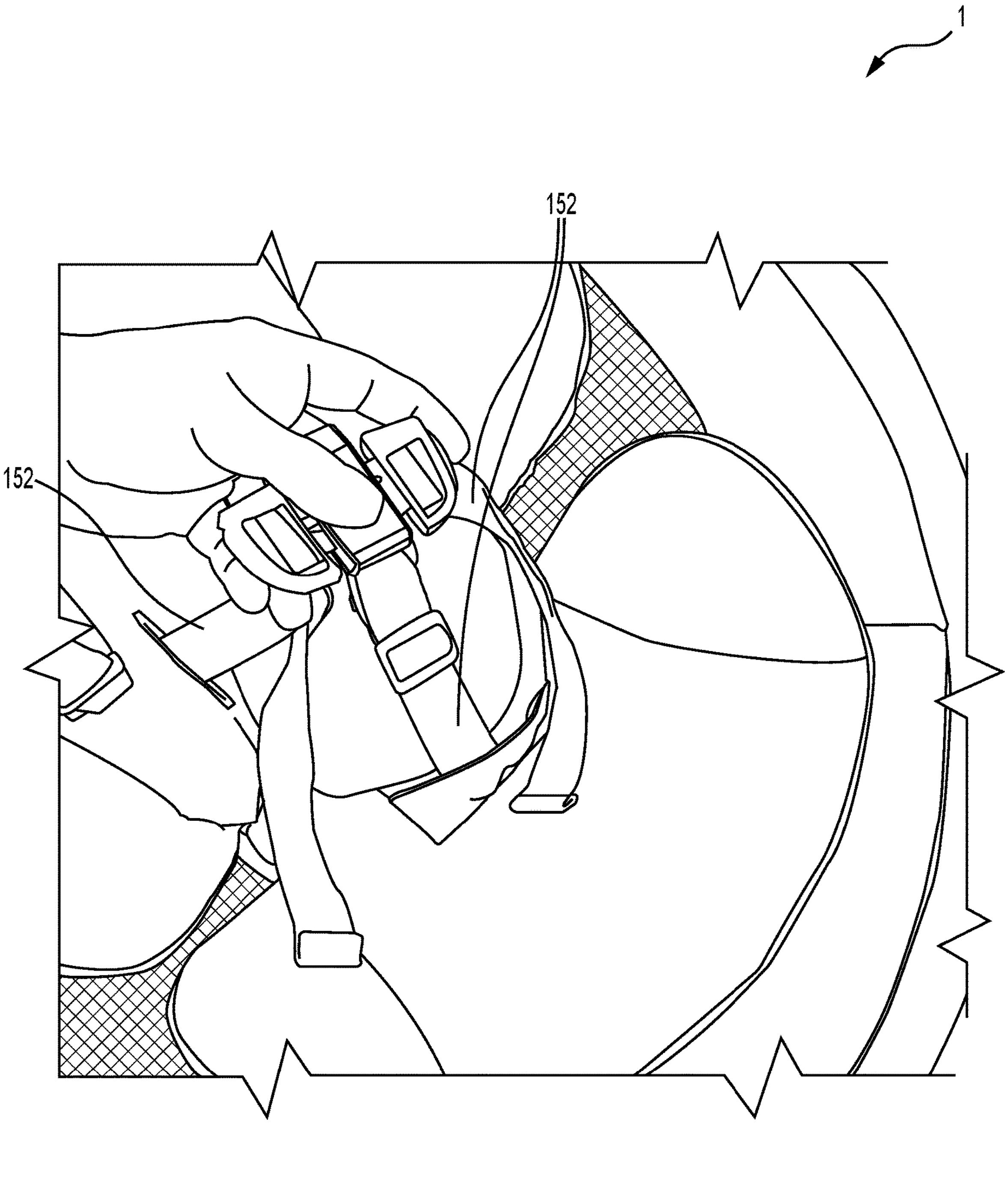

FIG. 17

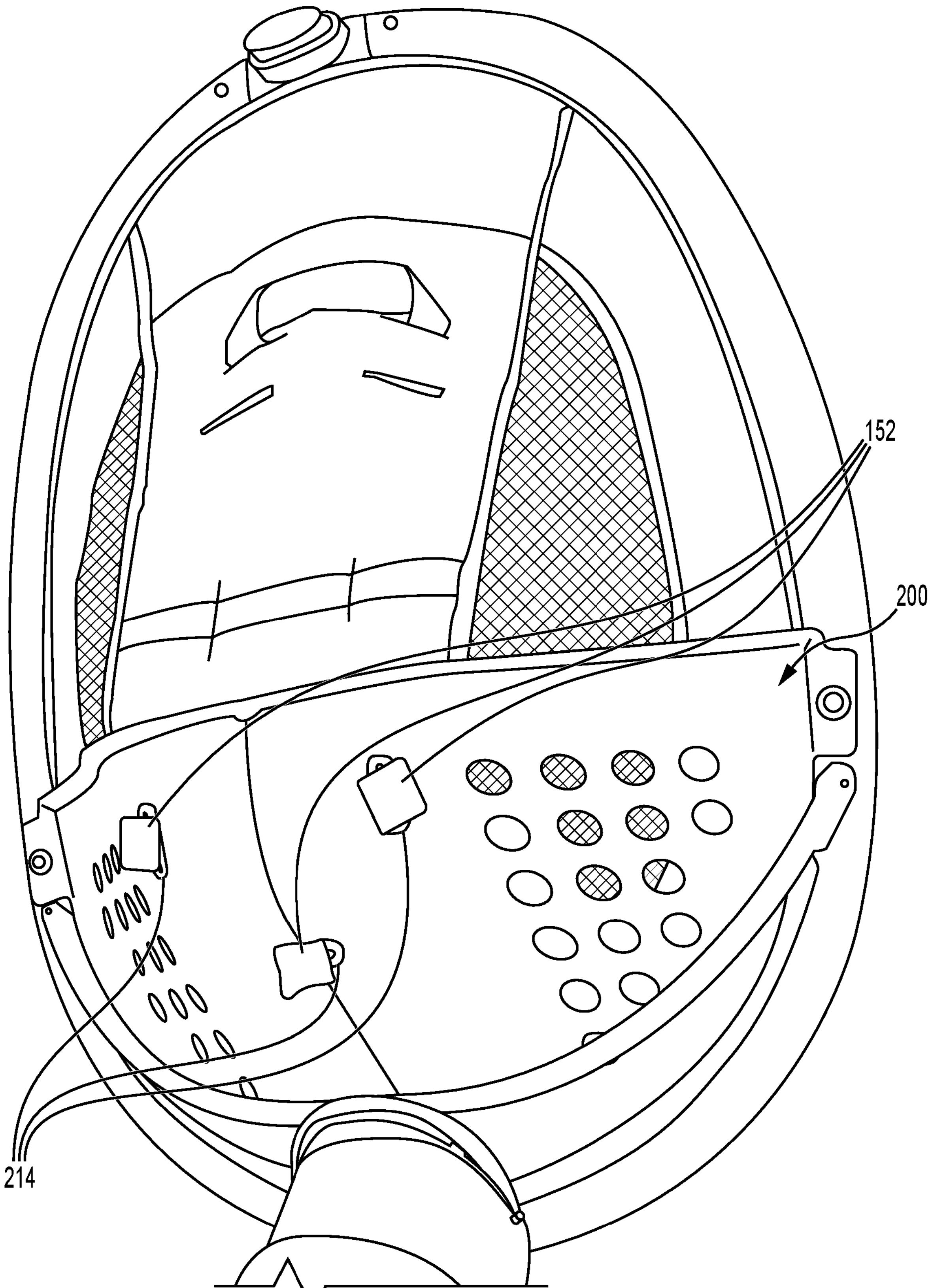

FIG. 18

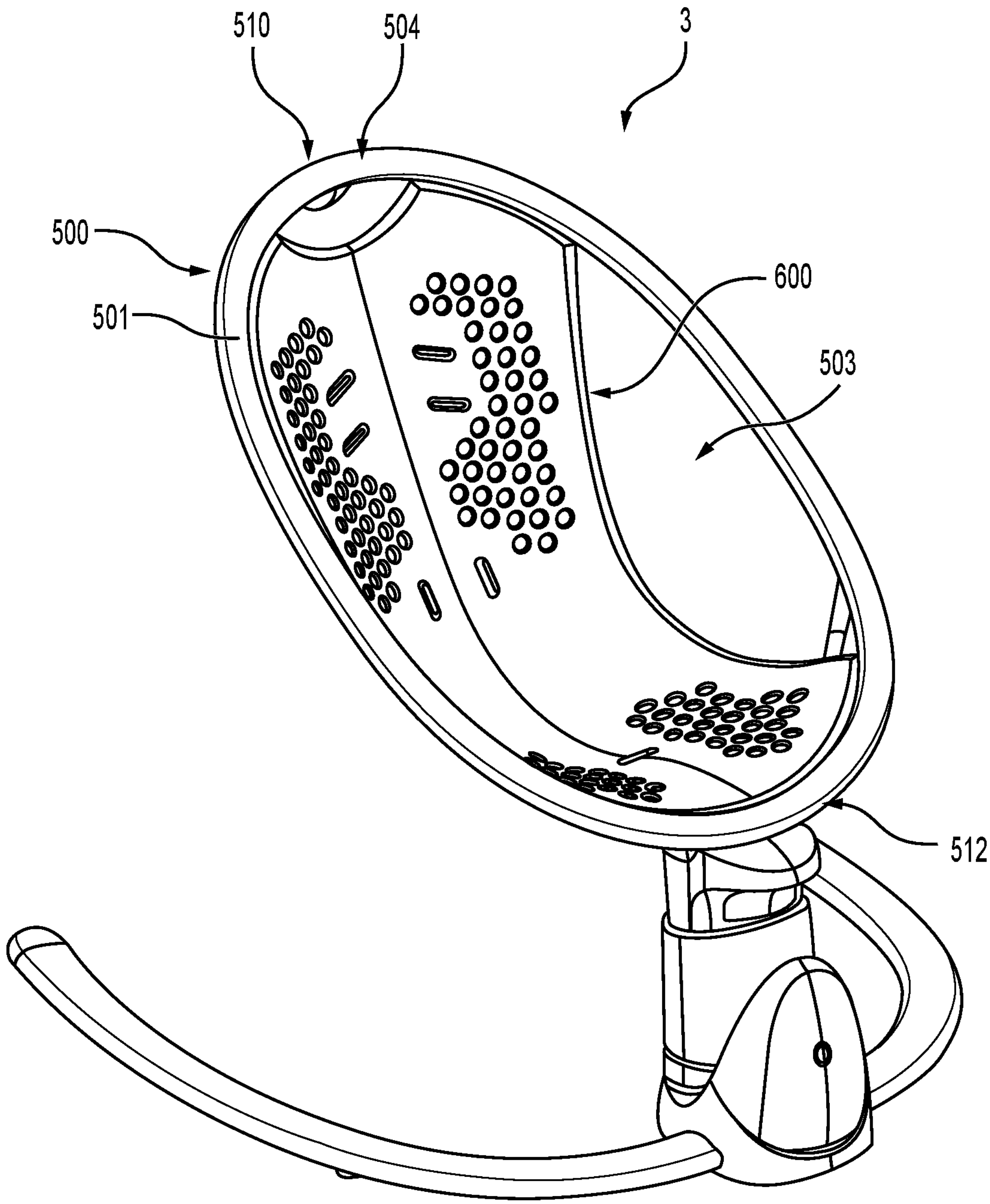

FIG. 19

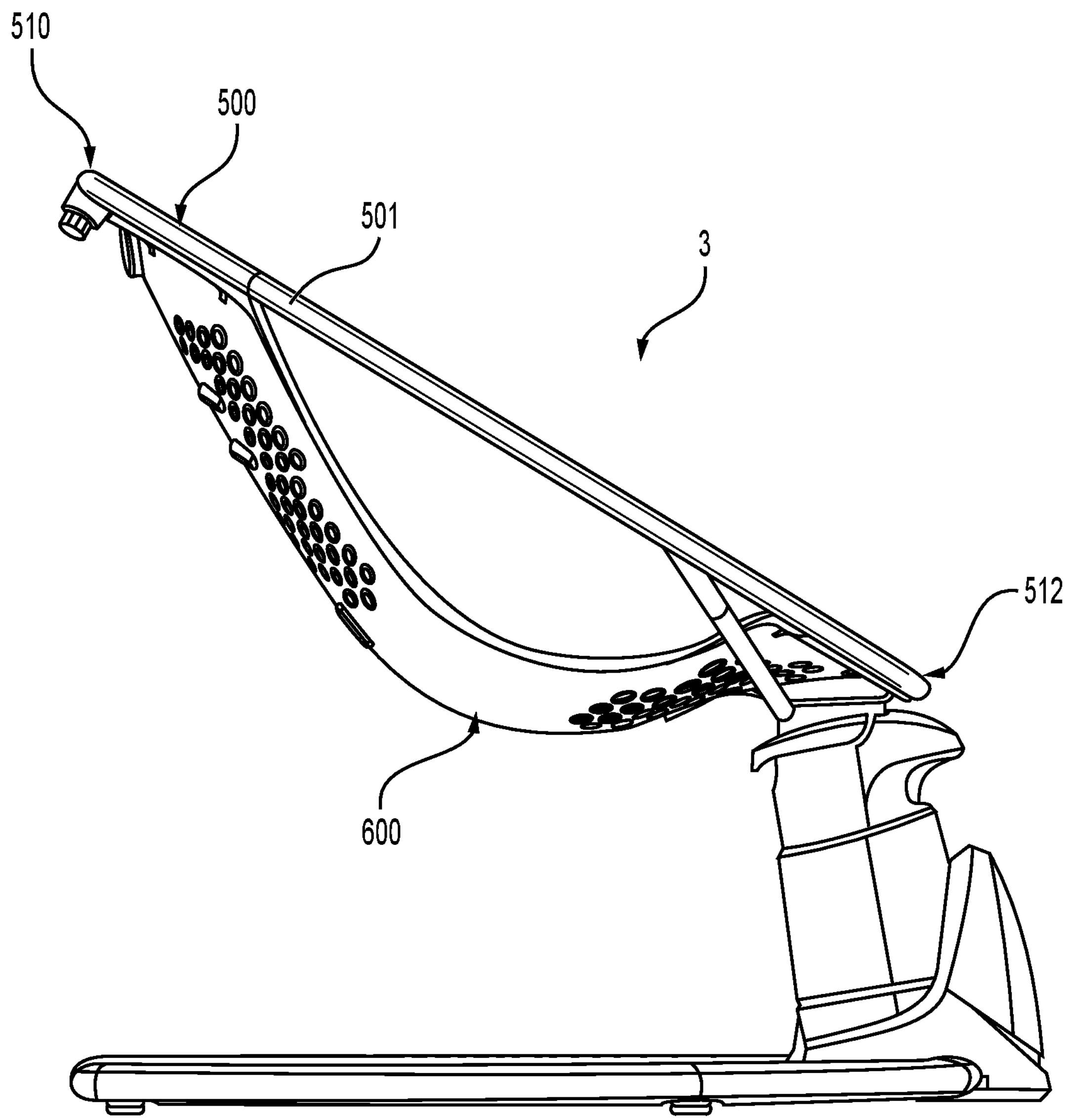

FIG. 20

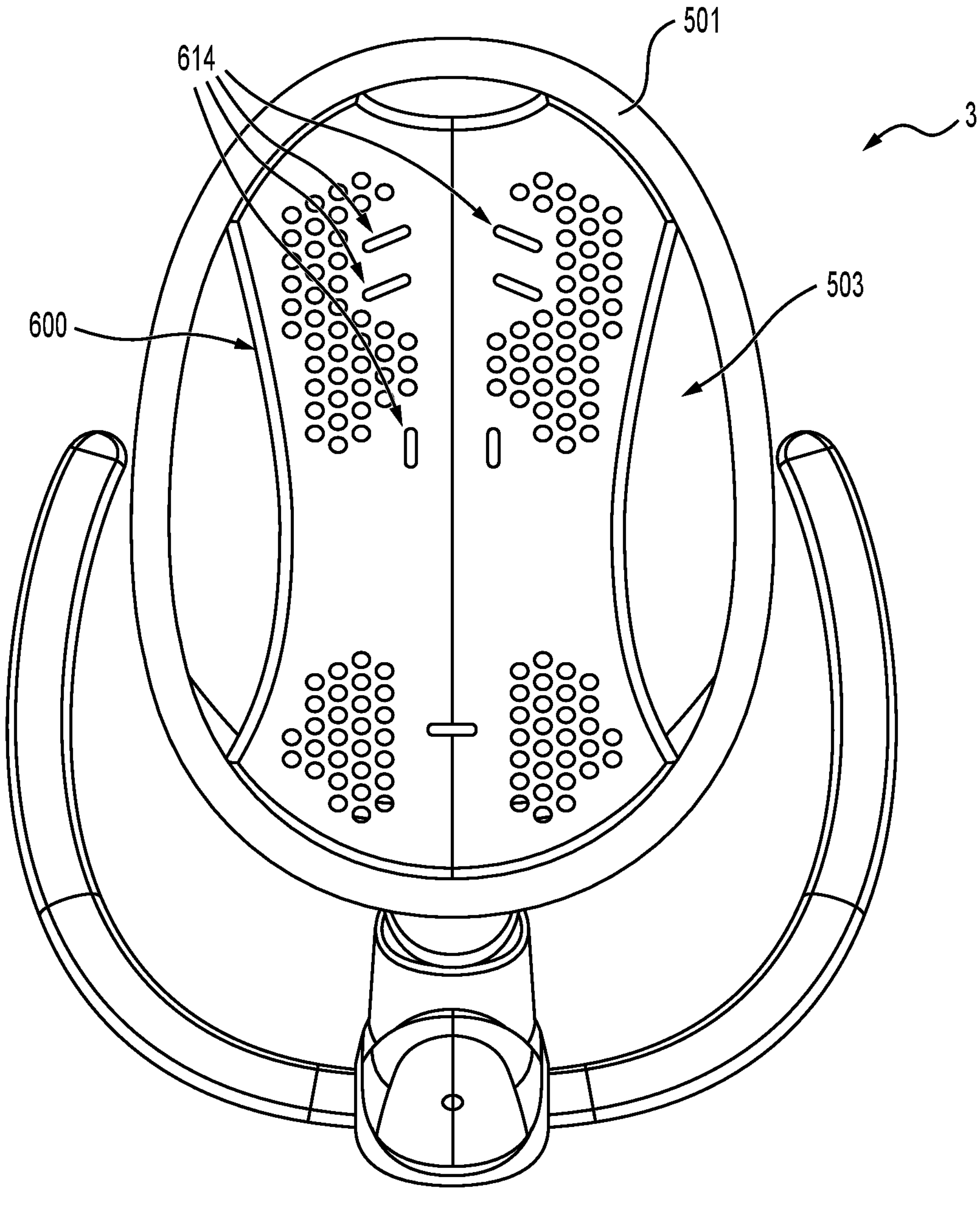

FIG. 21

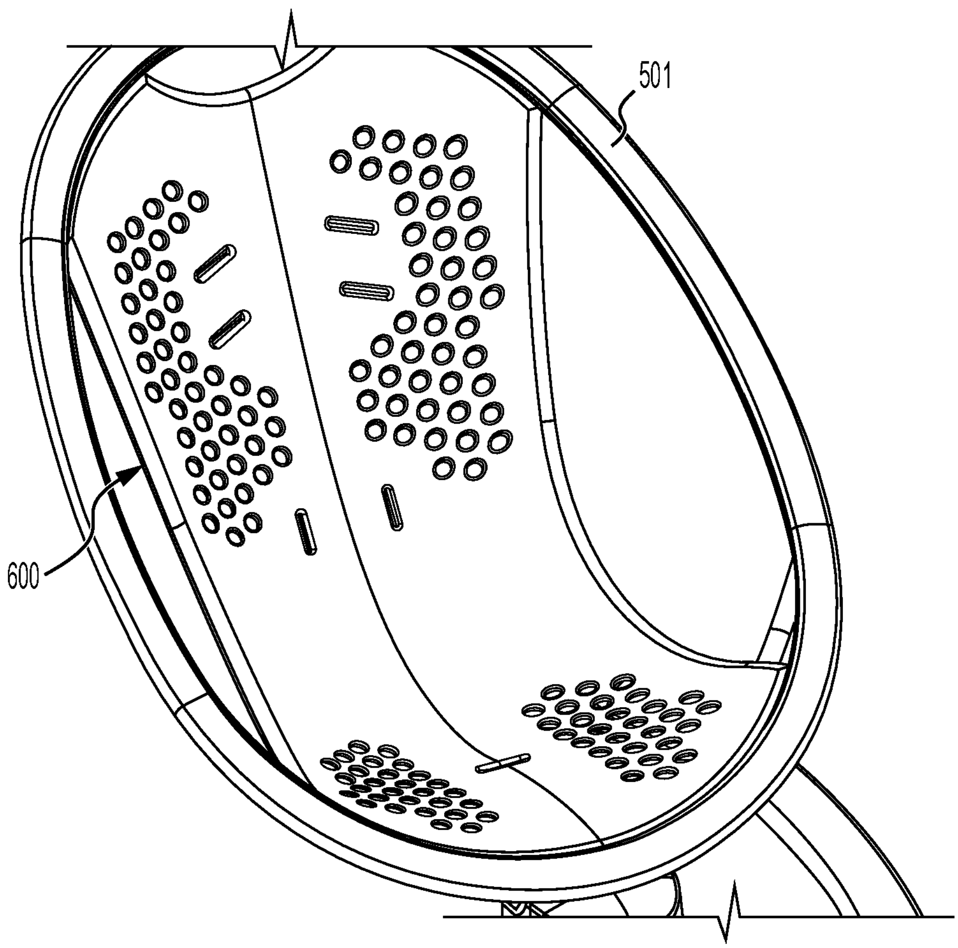

FIG. 22

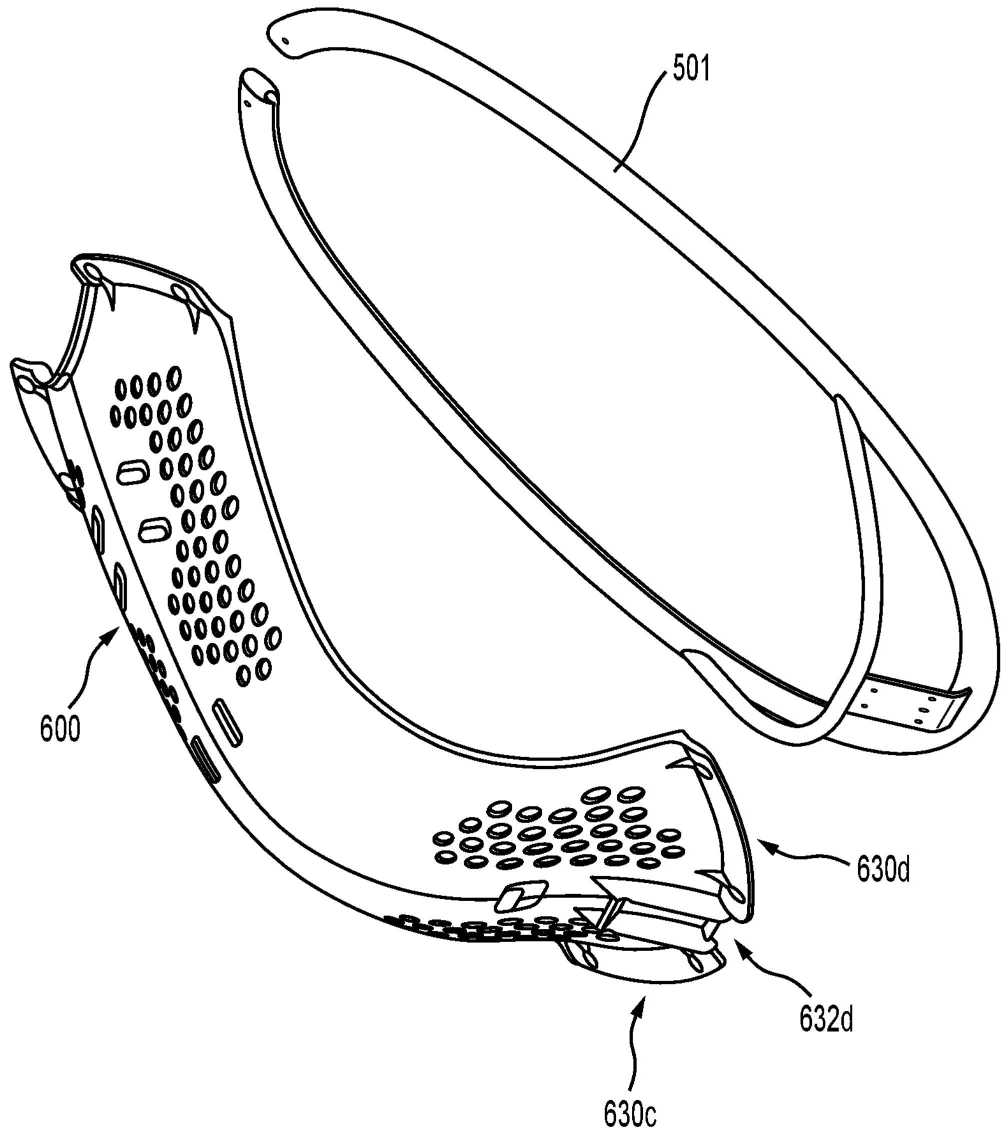

FIG. 23A

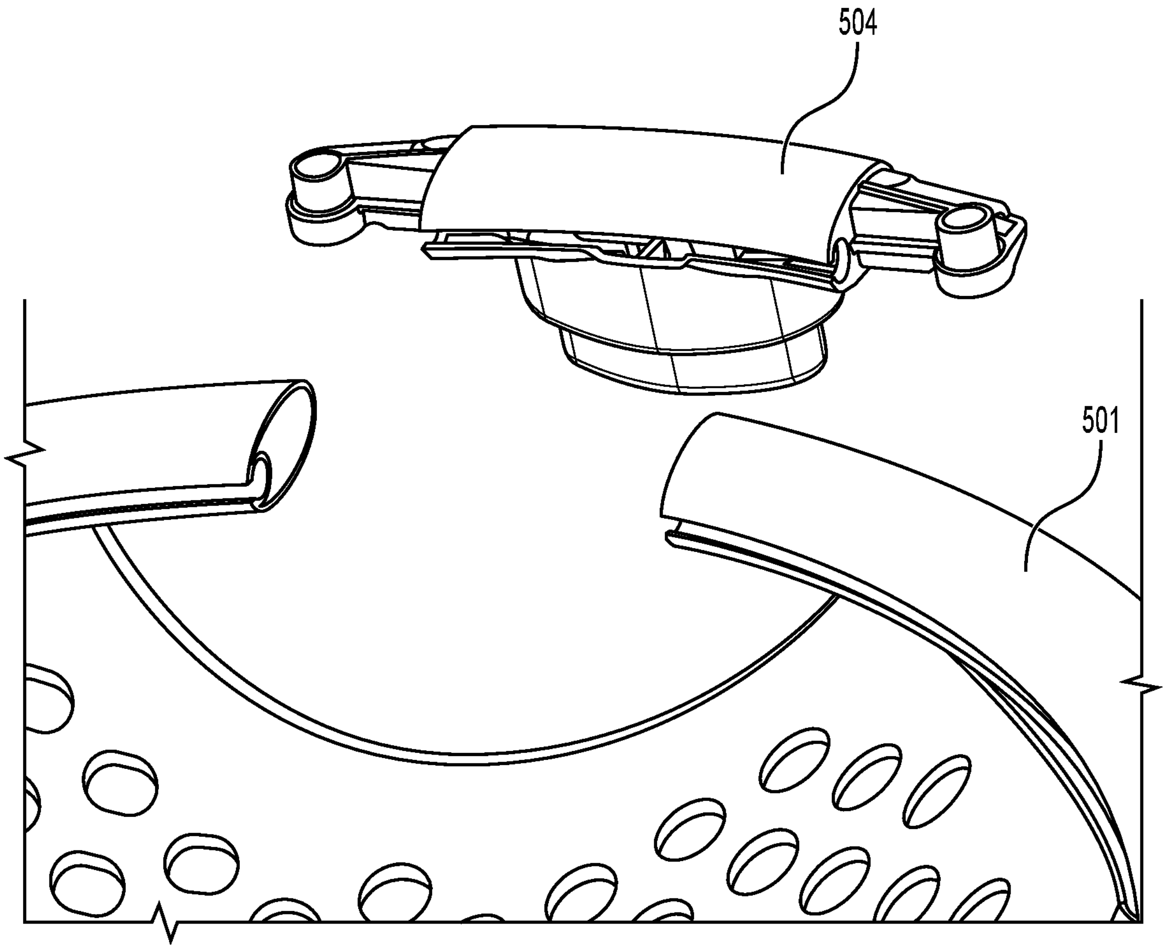

FIG. 23B

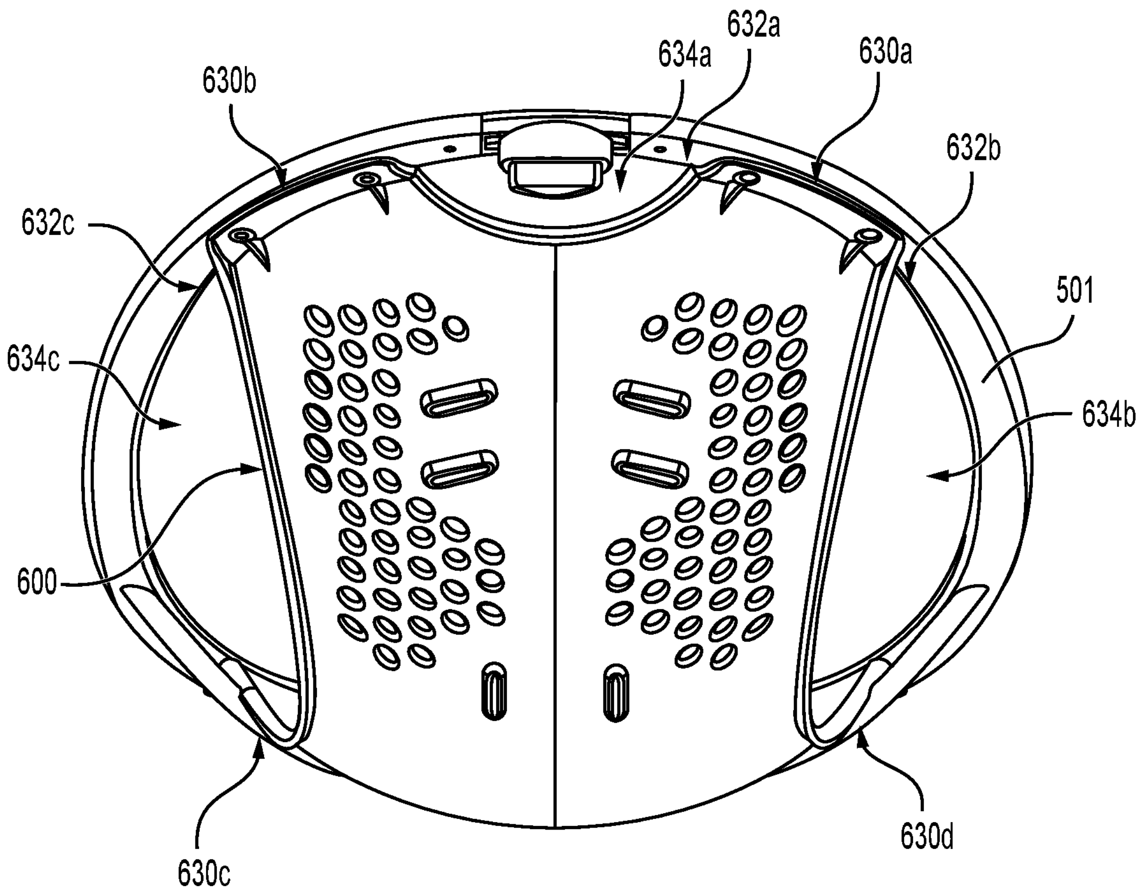

FIG. 24

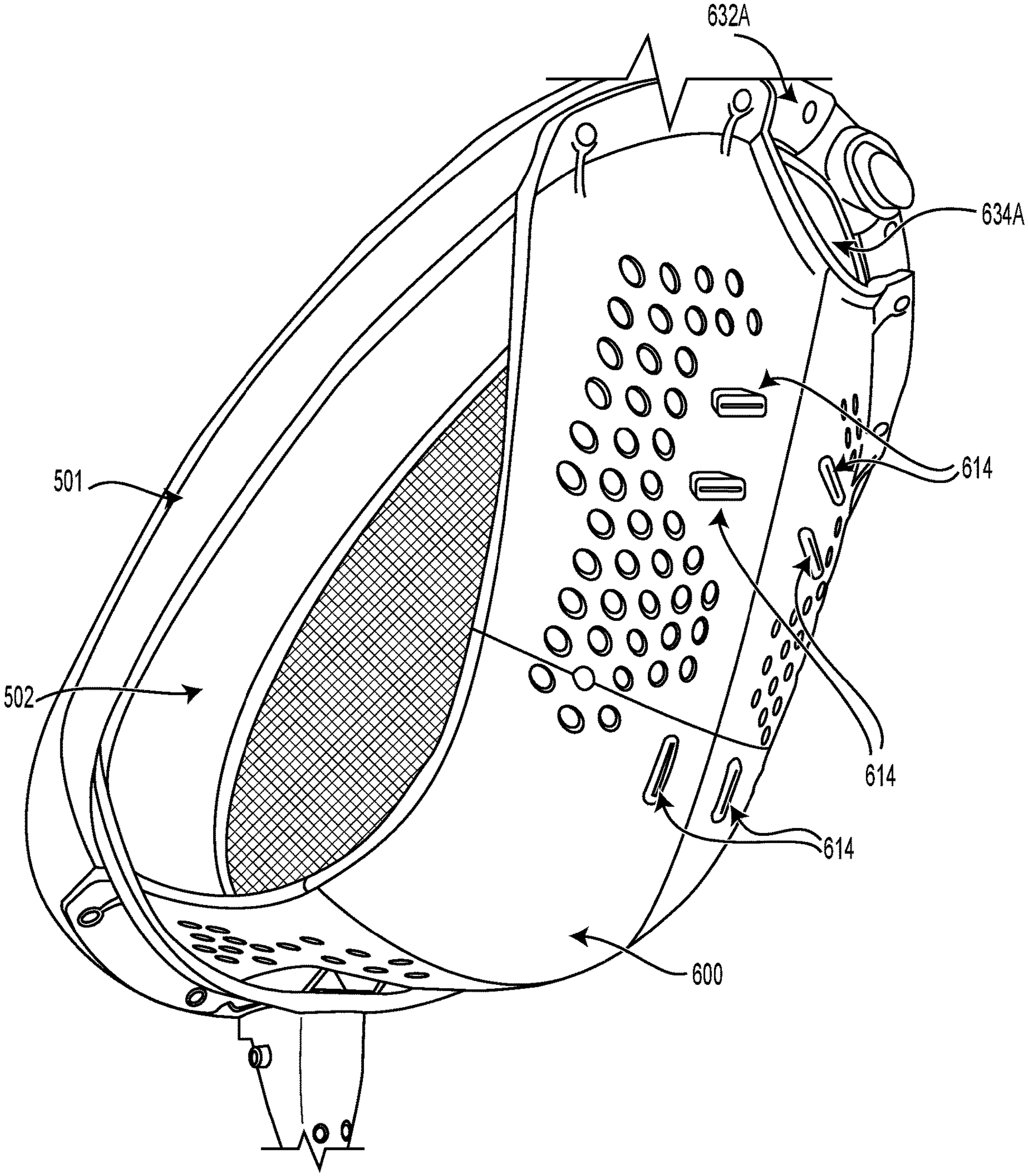

FIG. 25

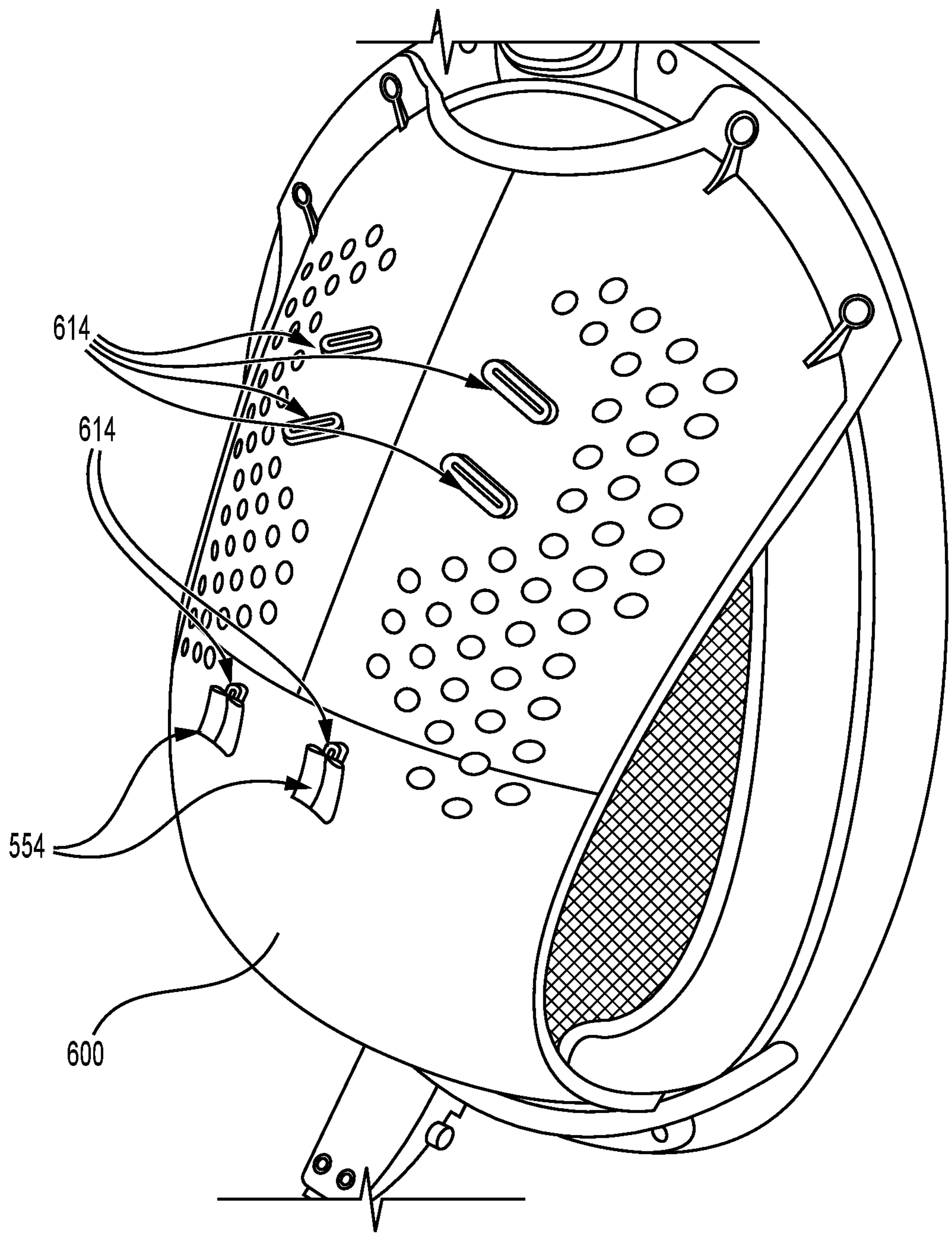

FIG. 26

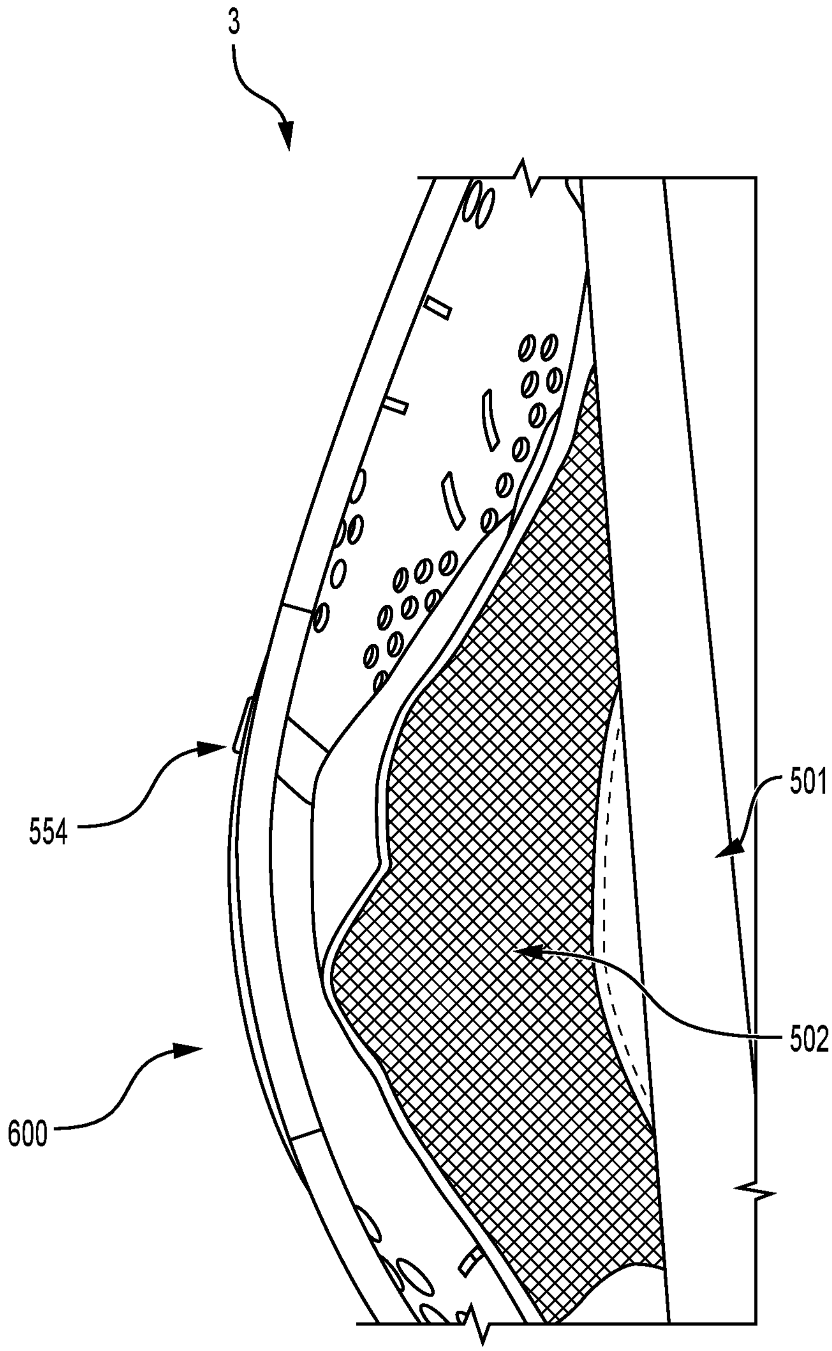

FIG. 27

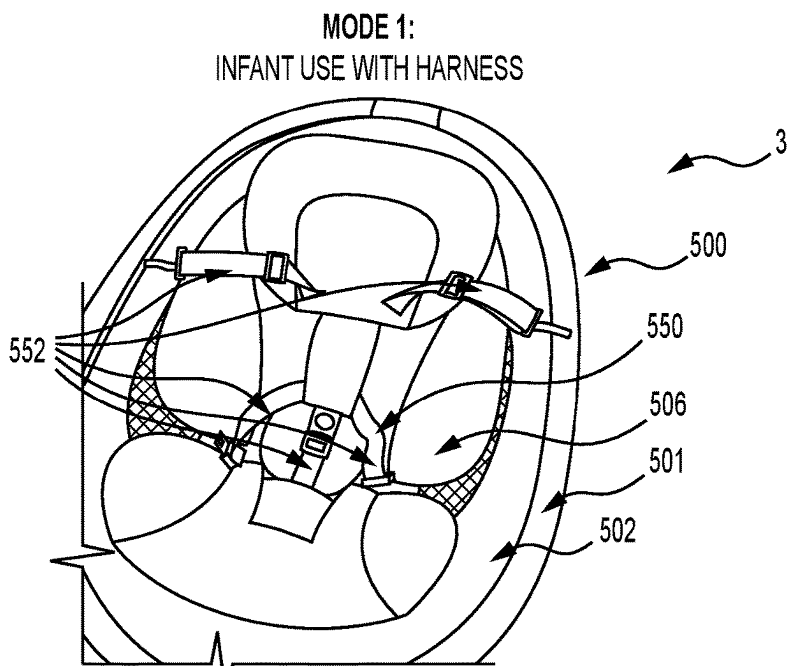

FIG. 28A

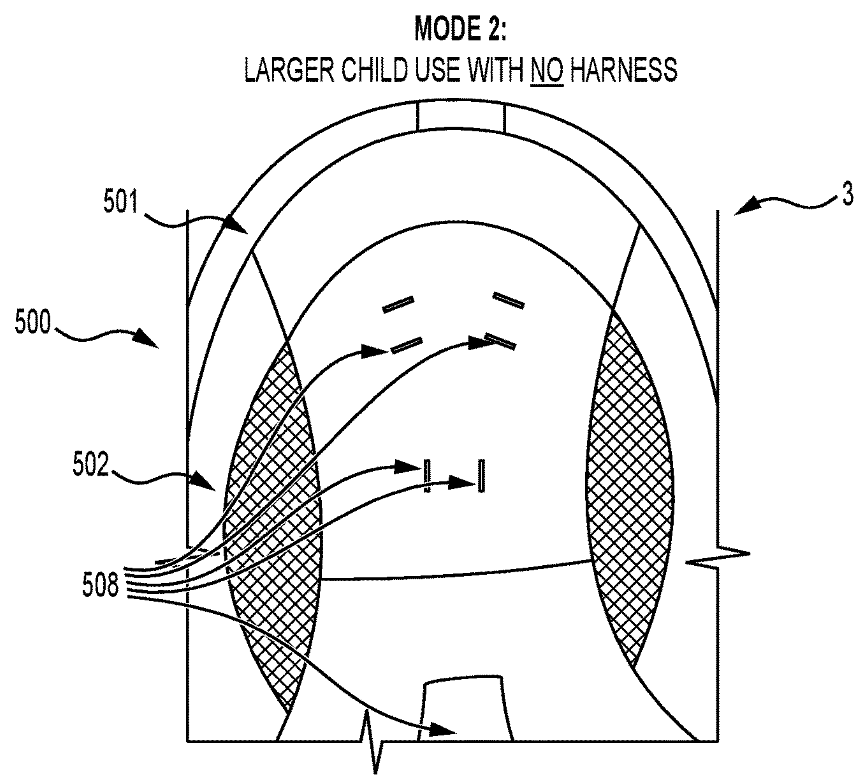

FIG. 28B

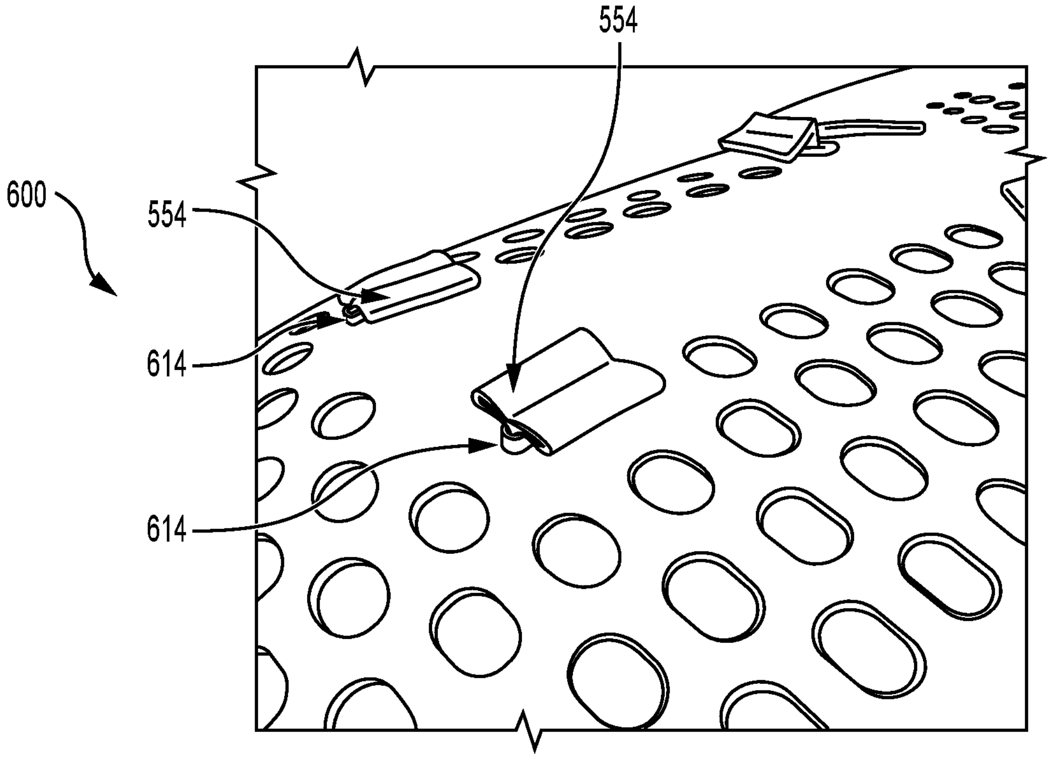

FIG. 29A

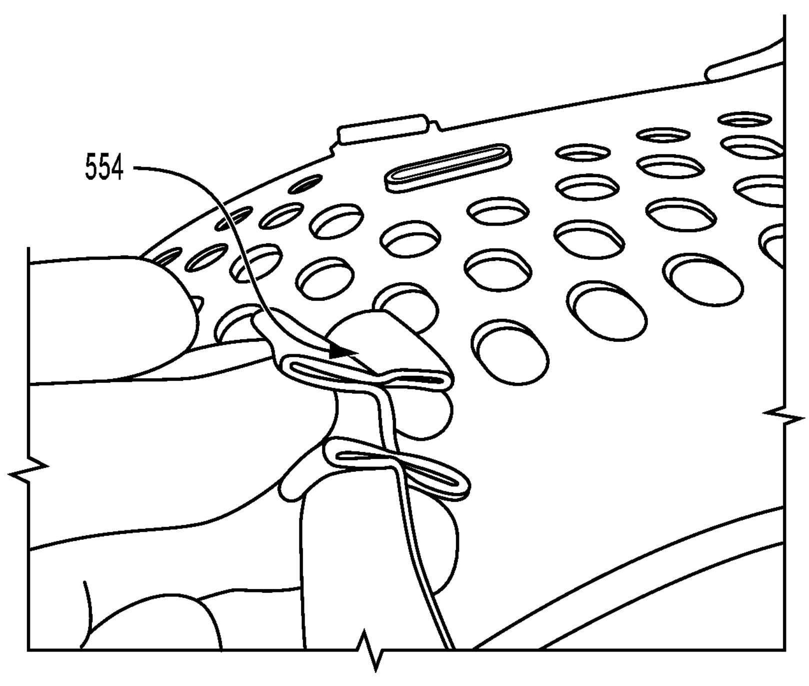

FIG. 29B

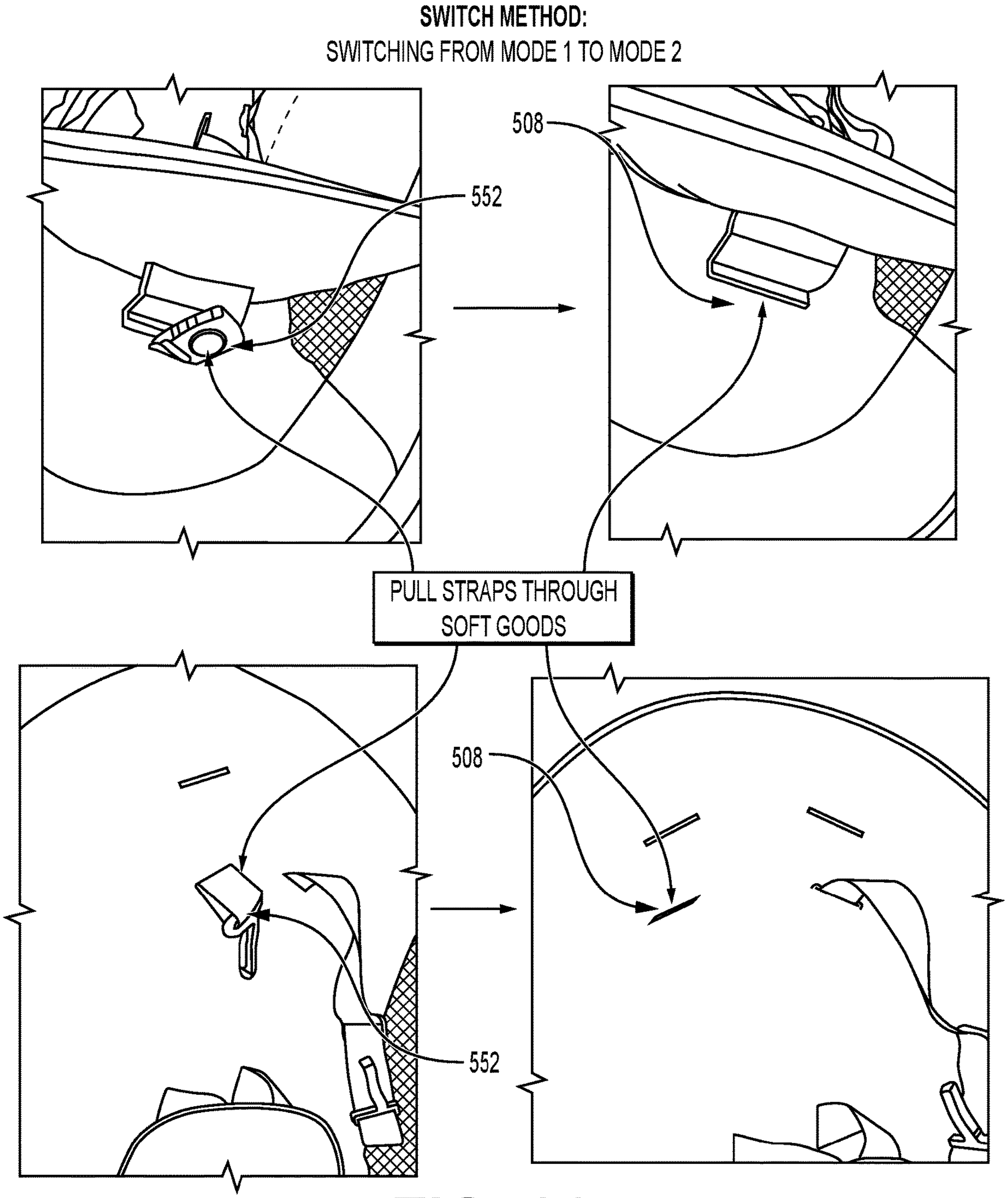
SWITCH METHOD:
SWITCHING FROM MODE 1 TO MODE 2
508
552
PULL STRAPS THROUGH SOFT GOODS
508
552

FIG. 30

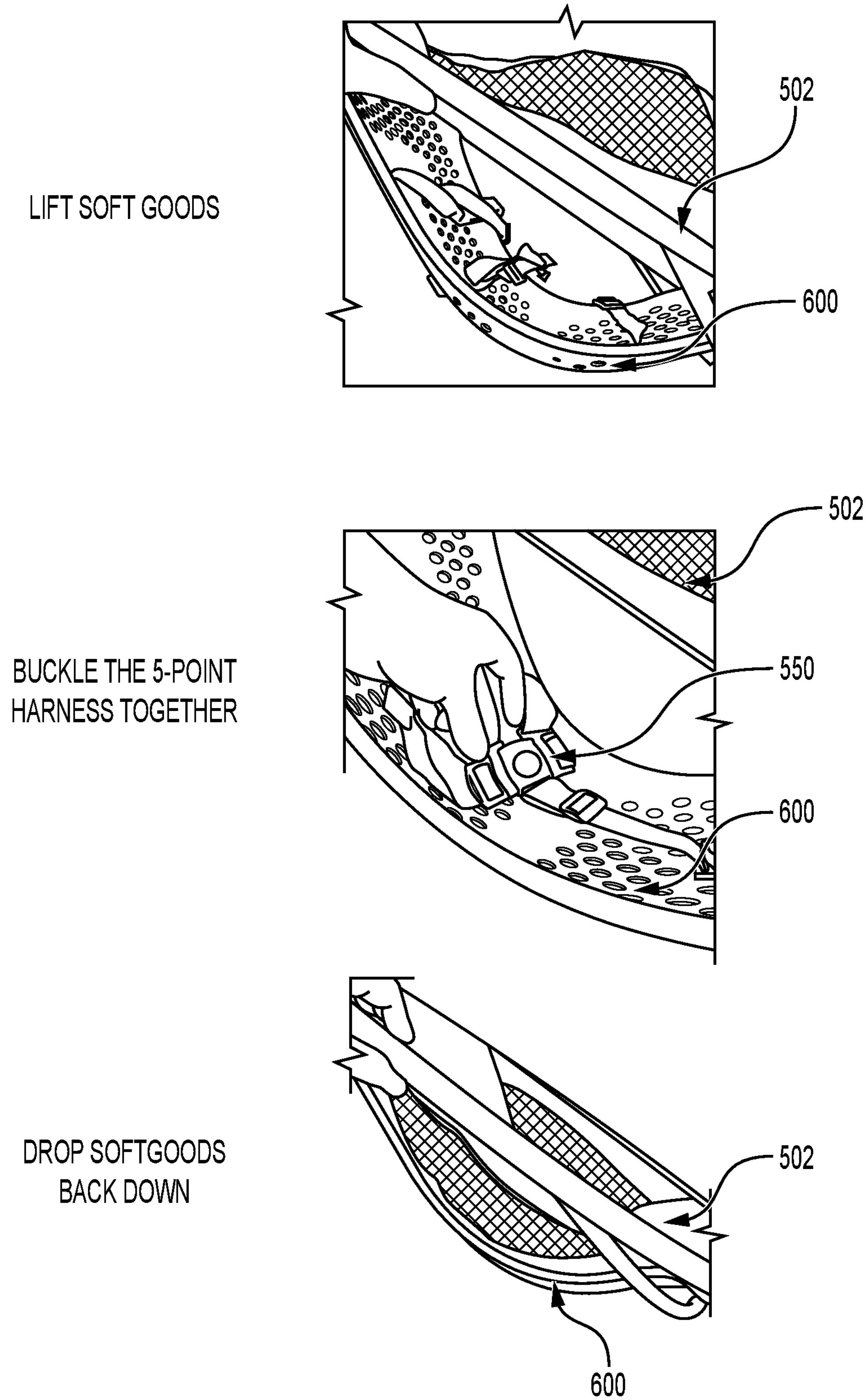
LIFT SOFT GOODS
502
600
BUCKLE THE 5-POINT
HARNESS TOGETHER
502
550
600
DROP SOFTGOODS
BACK DOWN
502
600

FIG. 31

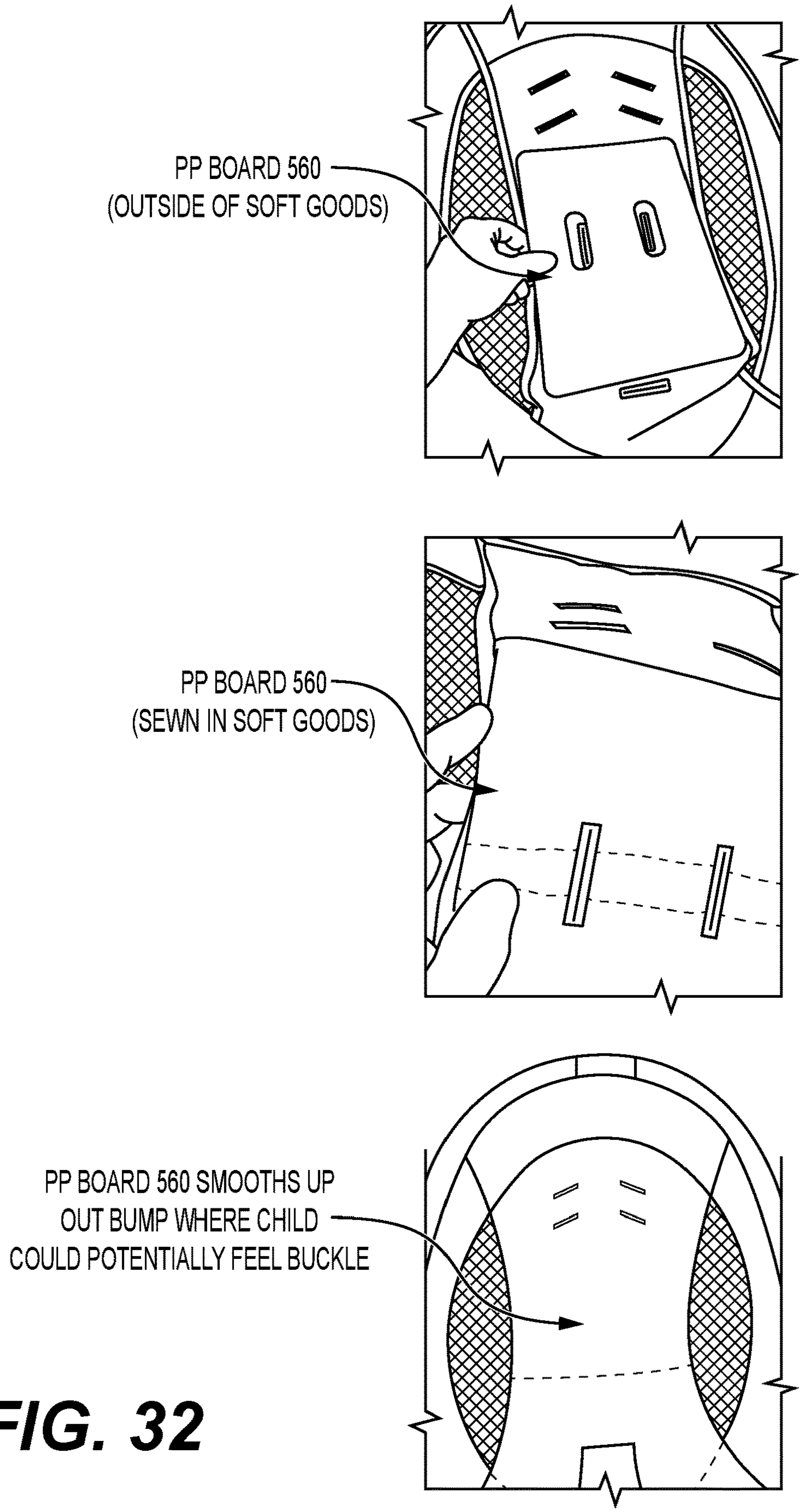
PP BOARD 560
(OUTSIDE OF SOFT GOODS)
PP BOARD 560
(SEWN IN SOFT GOODS)
PP BOARD 560 SMOOTHS UP
OUT BUMP WHERE CHILD
COULD POTENTIALLY FEEL BUCKLE

FIG. 32

SEAT SUPPORT FOR CHILD SWING

PRIORITY CLAIM

The present disclosure claims priority to U.S. Provisional Patent Application Ser. No. 63/369,802 filed Jul. 29, 2022; U.S. Provisional Patent Application Ser. No. 63/401,091 filed Aug. 25, 2022; and U.S. Provisional Patent Application Serial No. 63/414,217 filed Oct. 7, 2022. The disclosure of the above-mentioned applications/patents which are incorporated herewith by reference.

TECHNICAL FIELD

The present disclosure relates generally to child motion apparatuses and, in particular, to child swing apparatuses.

BACKGROUND

Infant swing apparatuses have become common household items. An infant swing has the primary function of applying a gentle motion, such as a swinging, rocking, or gliding motion, to soothe a child, while providing a safe and comfortable seating area. Infant swings are sold in various shapes, sizes, and configurations. A common style of infant swing includes a frame and an infant seat that oscillates relative to the frame. The swing arm moves to impart the motion to the infant seat.

The foregoing background discussion is intended solely to aid the reader. It is not intended to limit the innovations described herein. Thus, the foregoing discussion should not be taken to indicate that any particular element of a prior system is unsuitable for use with the innovations described herein, nor is it intended to indicate that any element is essential in implementing the innovations described herein.

SUMMARY

The present application discloses a child swing that comprises a base, a column, and a seat. The base is configured to support the child swing on a floor. The column extends upwards from the base and defines an axis of rotation. The seat is supported by the column above the base. The seat includes a seat rim and a seat support fixedly attached to the seat rim. The seat support is an integral part of the seat to provide additional support for a child positioned on the swing.

An aspect of the present disclosure provides a seat for a child swing. The child swing includes a base and a column extending from the base. The base is positionable on a floor. The seat comprises a seat rim and a seat support. The seat rim is connectable to the column to support the seat above the floor. The seat rim defines a seat opening. The seat support comprises a rigid material and is attached to the seat rim such that movement between the seat support and the seat rim is substantially prevented. The seat support defines a recess on one side of the seat opening. The seat support is positioned relative to the seat rim such that when a child is introduced through the seat opening from an opposing side of the seat opening, the seat support supports the child within the recess.

This summary is provided to introduce a selection of concepts in a simplified form that are further described below in the Detailed Description section. This Summary is not intended to identify key features or essential features of the claimed subject matter, nor is it intended to be used to limit the scope of the claimed subject matter. Furthermore, the claimed subject matter is not constrained to limitations that solve any or all disadvantages noted in any part of this disclosure.

BRIEF DESCRIPTION OF THE DRAWINGS

The foregoing summary, as well as the following detailed description of illustrative embodiments of the present application, will be better understood when read in conjunction with the appended drawings. For the purposes of illustrating the present application, there are shown in the drawings illustrative embodiments of the disclosure. It should be understood, however, that the application is not limited to the precise arrangements and instrumentalities shown. In the drawings:

FIG. **1** illustrates a front perspective view of a child swing, according to an aspect of this disclosure;

FIG. **2** illustrates a front perspective view of the child swing shown in FIG. **1** without a seating surface, according to an aspect of this disclosure;

FIG. **3** illustrates a side view of the child swing shown in FIG. **2**;

FIG. **4** illustrates a top side view of the child swing shown in FIG. **2**;

FIG. **5** illustrates a top perspective view of a portion of the child swing shown in FIG. **2**;

FIG. **6** illustrates a bottom perspective view of a portion of the child swing shown in FIG. **2**;

FIG. **7** illustrates an exploded view of a seat of the child swing shown in FIG. **2**;

FIG. **8** illustrates a rear side assembled view of the seat shown in FIG. **7**;

FIG. **9** illustrates a front perspective view of a child swing without a seating surface, according to another aspect of this disclosure;

FIG. **10** illustrates a side view of the child swing shown in FIG. **9**;

FIG. **11** illustrates a top perspective view of a portion of the child swing shown in FIG. **9**;

FIG. **12** illustrates a top side view of the child swing shown in FIG. **9**;

FIG. **13** illustrates a bottom perspective view of a portion of the child swing shown in FIG. **9**;

FIG. **14**A illustrates an exploded view of a seat of the child swing shown in FIG. **9**;

FIG. **14**B and **14**C illustrate perspective views an actuator connector, according to an aspect of this disclosure;

FIG. **15** illustrates a side assembled view of the seat shown in FIG. **14**A;

FIG. **16** illustrates a bottom perspective view of a portion of the child swing shown in FIG. **1**;

FIG. **17** illustrates a top perspective view of a portion of the child swing shown in

FIG. **1**;

FIG. **18** illustrates a bottom perspective view of a portion of the child swing shown in FIG. **1**;

FIG. **19** illustrates a front perspective view of a child swing according to another aspect of this disclosure;

FIG. **20** illustrates a side view of the child swing shown in FIG. **19**;

FIG. **21** illustrates a top side view of the child swing shown in FIG. **19**;

FIG. **22** illustrates a top perspective view of a portion of the child swing shown in FIG. **19**;

FIG. **23**A illustrates an exploded view of a seat of the child swing shown in FIG. **19**;

FIG. **23**B illustrates an actuator connector of the seat of the child swing shown in FIG. **19**;

FIG. **24** illustrates a bottom perspective view of the seat of the child swing shown in FIG. **19**;

FIG. **25** illustrates a bottom perspective view of the child swing shown in FIG. **19**;

FIG. **26** illustrates a bottom perspective view of the child swing shown in FIG. **19**;

FIG. **27** illustrates a side view of a portion of the child swing shown in FIG. **19**;

FIG. **28**A illustrates a child swing in an infant mode, according to an aspect of this disclosure;

FIG. **28**B illustrates a child swing in child mode, according to an aspect of this disclosure;

FIGS. **29**A and **29**B illustrate a connection between a restraint assembly and a child seat, according to an aspect of this disclosure;

FIGS. **30** and **31** illustrate a method of removing a restraint assembly from soft goods, according to an aspect of this disclosure; and

FIG. **32** illustrates a child seat having soft goods with a support board, according to an aspect of this disclosure.

DETAILED DESCRIPTION

Certain terminology used in this description is for convenience only and is not limiting. The words "top", "bottom", "upper", "lower", "above", "below", "axial", "transverse", "circumferential," and "radial" designate directions in the drawings to which reference is made. The term "substantially" and derivatives thereof, and words of similar import, when used to describe sizes, shapes, spatial relationships, distances, directions, and other similar parameters includes the stated parameter in addition to a range up to 10% more and up to 10% less than the stated parameter, including 5% more and 5% less, including 3% more and 3% less, including 1% more and 1% less. All ranges disclosed herein are inclusive of the recited endpoint and independently combinable (for example, the range of "from 2 grams to 10 grams" is inclusive of the endpoints, 2 grams and 10 grams, and all the intermediate values). The terminology includes the above-listed words, derivatives thereof and words of similar import.

FIGS. **1-8** show a child swing **1** according to an aspect of this disclosure. FIG. **1** illustrates a front perspective view of the child swing **1**. The child swing **1** comprises a base **10**, a column **20**, and a seat **100**. The column **20** is positioned between the base **10** and the seat **100**. The base **10** supports the seat **100** above a support surface such as a floor. The seat **100** is configured to move by, for example, swinging, rocking, or gliding relative to the base **10**. The column **20** can comprise an extendable column configured to adjust a height of the seat **100** above the support surface. The child swing **1** can further include, for example, a recline mechanism **40** (see FIGS. **2** and **3**), a magnetic drive (not shown), a seat motion sensor (not shown), legs **80***a*, **80***b*, a vibration device (not shown), or still other features. However, it will be understood that alternative child swings of this disclosure need not be implemented with all of the recline mechanism **40**, the magnetic drive, the seat motion sensor, the legs **80***a*, **80***b* and the vibration device. Rather, alternative child swings of this disclosure can be implemented with fewer than, or include more than, all of the recline mechanism **40**, the magnetic drive, the seat motion sensor, the legs **80***a*, **80***b*, and the vibration device. For example, alternative child swings of this disclosure can include one or more, or any

combination of two or more, of the recline mechanism **40**, the magnetic drive, the seat motion sensor, the legs **80***a*, **80***b*, or the vibration device.

The base **10** is configured to support the child swing **1** on a floor or other surface. The column **20** extends upwards from the base **10** and can define an axis of rotation A_R (See FIG. **3**). The seat **100** can be configured to rotate about the axis of rotation A_R relative to the base **10**. The seat **100** is supported by the column **20** above the base **10**. For example, the column **20** can be attached to the seat **100** such that the seat **100** is disposed on top of the column **20**. The column **20** can be configured to transition the seat **100** between a plurality of height positions. For example, the column **20** is configured to transition the seat **100** between a lowered position in which the seat **100** is positioned at a first height above the floor, and a raised position in which the seat **100** is positioned at a second height above the floor greater than the first height.

The seat **100** comprises a seat rim **101** and a seating surface **102** that is configured to support a child thereon. The seating surface **102** can be a soft seating surface formed from soft goods that are suspended from the seat rim **101**. In some examples, the seat rim **101** can define a receiving channel **109** that extends around an inner perimeter of the seat rim **101**. The receiving channel **109** can be configured to receive an outer edge of the soft goods such that the seat rim **101** remains exposed (i.e., not covered by the soft goods) when the soft goods is attached to the seat rim **101**. The seating surface **102** can also include additional soft goods attached to the seating surface **102**. For example, the seating surface **102** can include a seating insert **104** positioned on top of the seating surface **102**. The seating insert **104** can comprise a seatback **106** and a seat pan **108** formed from soft goods. The seating insert **104** can provide additional support and padding for a child.

In an alternative aspect, the seating surface **102** can be a rigid seating surface formed from a rigid material, such as a polymer and/or a plastic, that defines the seat rim **101**. The rigid seating surface can be covered in soft goods to provide cushioning for the child.

FIGS. **2-4** illustrate the child swing **1** without the seating surface **102** attached to the seat rim **101** according to an aspect of this disclosure. FIG. **2** illustrates a front perspective view of the child swing **1**; FIG. **3** illustrates a side view of the child swing **1**; FIG. **4** illustrates a top side view of the child swing **1**. The seat rim **101** has an upper end **110** and an opposing lower end **112**, as shown in FIG. **3**. The seat rim **101** can have a ring shape or another suitable shape that extends between the upper and lower ends **110** and **112**. The seat rim **101** can be defined by a tubular ring or other suitable structure. The tubular ring can comprise a metal or other suitably rigid material. The seat rim **101** defines a seat opening **103** that extends along a seat rim plane, as shown in FIGS. **2** and **4**. The seat rim plane can be angularly offset from the axis of rotation A_R. The axis of rotation A_R can extend through the seat rim plane.

The seat **100** further includes a support member **114** and a seat support **200**, as shown in FIG. **3**. The support member **114** is configured to support the seat rim **101** on the column when the seat **100** is attached to the column **20**. For example, the seat **100** can be cantilevered from the column **20**. The seat rim **101** can be attached to the column **20** at the lower end **112** of the seat rim **101**. The support member **114** can be attached between the seat rim **101** and the column **20**. The support member **114** can extend from a location on the seat rim **101** between the upper end **110** and the lower end **112** to the column **20**.

It will be appreciated that in alternative examples, the seat **100** can be attached to the column **20** at another location on the seat rim **101**, such as a middle portion or the upper end **110** of the seat rim **101**. Similarly, the support member **114** can be attached between the seat rim **101** and the column **20** at various locations to support the seat **100** on the column **20**.

The recline mechanism **40** can be positioned between the seat **100** and the column as shown in FIGS. **2-3**. For example, the recline mechanism **40** can couple the seat **100** to the column **20**. The recline mechanism **40** can be attached to an upper end of the column **20**, and the seat rim **101** can be attached to the recline mechanism **40**. The support member **114** can extend from the seat rim **101** and be attached to the recline mechanism **40**. The recline mechanism **40** is configured to selectively transition the seat **100** between the plurality of recline positions. In an aspect, the seat **100** and the recline mechanism are rotatable about the axis of rotation A_R relative to the base **10**.

The seat **100** can be configured to removably couple to the column **20**. The seat **100** can be configured to couple to the column **20** such that the seat **100** is rotationally fixed relative to the column **20**, such that rotation of the column **20** causes a corresponding rotation of the seat **100**. Similarly, the recline mechanism **40** can be rotationally fixed relative to the column **20** such that rotation of the column **20** causes a corresponding rotation of the recline mechanism **40**.

The column **20** can comprise a shaft (not shown), which can be referred to as a pivot shaft, that defines the axis of rotation A_R. The axis of rotation A_R can define an angle relative to the floor. The angle can be **90** degrees. However, preferably, the angle is less than **90** degrees, as shown in FIG. **3**. Thus, the axis of rotation A_R can extend rearward as it extends upward away from the floor. Angling the axis of rotation A_R in such a manner can cause the seat **100** to sway in a manner that mimics a natural pendulum. It will be appreciated that the column **20** can include one or more shafts that can, for example, (1) define the axis of rotation A_R and (2) extend and retract (e.g., a telescope) to adjust a height of the seat **100**.

FIG. **5** illustrates a top perspective view of a portion of the child swing **1** including the seat support **200**. The seat support **200** is attached to the seat rim **101**. The seat support **200** can be attached to the seat rim **101** via one or more fasteners **120** to create a rigid connection. It will be appreciated that the one or more fasteners can include, for example, screws, bolts, pins, pop rivets, or still other types of fasteners. It will be further appreciated that the seat support **200** can be connected to the seat rim **101** using glue or other adhesives, a snap-fit, a compression fit, combinations thereof, combinations of fasteners, fits, and adhesives, or still other connection types or methods. The seat support **200** is positioned relative to the seat rim **101** such that when a child is introduced through the seat opening **103** from above the child swing **1**, the seat support **200** supports the child.

FIG. **6** illustrates a bottom perspective view of a portion of the child swing **1** including the seat support **200**; FIG. **7** illustrates an exploded view of the seat **100** of the child swing **1**. FIG. **8** illustrates a rear side assembled view of the seat **100**. The seat support **200** can be attached to an underside surface **105** of the seat rim **101**. The seat surface **102** can be attached to the seat rim **101** within a groove **107** defined by the seat rim **101**, as shown in FIG. **7**. The groove **107** can extend about the seat opening **103**. In an aspect, the underside surface **105** is positioned between the groove **107** and the seat support **200**. The attachment of the seat support **200** to the seat rim **101** allows the seat surface **102** to be attached to the seat rim **101** without interfering with the attachment of the seat support **200**.

The seat support **200** can be attached to the seat rim **101** at one location or more than one location on the underside surface **105**. The seat support **200** can contact the seat rim **101** along a portion of a length of the seat rim **101** (e.g., a contact portion). The length of the seat rim **101** can extend about the entire seat opening **103**. In an aspect, the contact portion is less than half of a length of the seat rim. The contact between the seat support **200** and the seat rim **101** can be continuous along the contact portion. Alternatively, the contact between the seat support **200** and the seat rim **101** along the contact portion can include breaks. For example, the seat support **200** can contact the seat rim **101** at three locations along the contact portion. Each of the three locations can be a location in which a fastener **120** connects the seat support **200** to the seat rim **101**. It will be appreciated that the number of breaks along the contact portion can include fewer or more than three breaks.

With reference to FIG. **6**, a contact portion **230** between the seat rim **101** and the seat support **200** includes a break **232**. The break **232** can be defined by a location along an edge of the seat support **200** that is not in contact with the seat rim **101** and is adjacent to the contact portion **230** between the seat support **200** and the seat rim **101** along the length of the seat rim **101**.

The seat support **200** can comprise a rigid material. In an aspect, the rigid material can include a plastic, for example, a polyethylene terephthalate (PET), a high-density polyethylene (HDPE), a polyvinyl chloride (PVC), a low-density polyethylene (LDPE), a polypropylene (PP), a polystyrene (PS), or still other type of plastic. The rigid material of the seat support **200** and the rigid attachment between the seat support **200** and the seat rim **101** substantially prevents movement between the seat support **200** and the seat rim **101**. For example, the seat support **200** and seat rim **101** combination can be integrally formed together to form a single substantially rigid part. The integrally formed seat support **200** and seat rim **101** combination provides additional support to the seat **100** so that both the seating surface **102** and the seat support **200** support a child seated on the seat **100**. In an aspect, the seat rim **101** comprises a metal and the seat support comprises a plastic.

With reference to FIG. **4**, the angle of the top view of FIG. **4** is in a direction that is substantially perpendicular to the seat rim plane. The seat support **200** can be sized such that when the seat **100** is viewed in the direction substantially perpendicular to the seat rim plane (view in FIG. **4**), an area of the seat support **200** is less than 50% of an area of the seat opening **103**. Stated another way, when the seat **100** is viewed from the direction substantially perpendicular to the seat rim plane, the seat support **200** would be visible in less than 50% of the seat opening **103** area. In an alternative aspect, seat support **200** can be sized such that when the seat **100** is viewed in the direction substantially perpendicular to the seat rim plane, an area of the seat support **200** is less than 40% of an area of the seat opening **103**. In another alternative aspect, the seat support **200** can be sized such that when the seat **100** is viewed in the direction substantially perpendicular to the seat rim plane, an area of the seat support **200** is less than 30% of an area of the seat opening **103**. The size of the seat support **200** can be selected to, for example, minimize the overall weight of the seat **100** while still providing support to a child positioned on the seat **100**.

With reference to FIGS. **5** and **8**, the seat support **200** has an inner surface **202** that defines a recess **210**. The recess **210** is defined on one side (e.g., a bottom side) of the seat opening **103**. The position of the seat support **200** and the recess **210** relative to the seat rim **101** is such that when a child is introduced through the seat opening **103** from an opposing side (e.g., a top side) of the seat opening **103**, the seat support **200** supports the child within the recess **210**. It will be appreciated that when the seating surface **102** is attached to the seat rim **101**, the seating surface **102** can also be supported by the seat support **200**, at least partially, when the child is introduced through the seat opening **103** and onto the seating surface **102**.

The inner surface **202** of the seat support **200** can define a contoured shape. The contoured shape of the inner surface **202** can be substantially symmetric relative to a plain substantially perpendicular to the seat opening **103** that extends from the upper end **110** to the lower end **112** of the seat rim **101**. The contoured shape of the inner surface **202** can help keep the child centered, to provide better balance for swing motion of the seat **100**.

The seat support **200** can comprise at least one vent hole **212** and at least one restraint hole **214**. The vent hole **212** can extend from the inner surface **202** to an outer surface **204** of the seat support **200**, to connect the recess **210** to an exterior of the seat **100**. The vent hole(s) **212** can be spaced about the seat support **200**. In an aspect, each of the vent holes **212** can have a substantially similar size and shape as each of the other vent holes. The vent holes **212** can allow air access through the seat support **200** when a child is supported thereon to provide a cooling affect.

With reference to FIG. **1**, the seat **100** further comprises a child restraint assembly **150**. The restraint assembly **150** can include one or more restraint straps **152**. The restraint straps **152** can be connected to the seat **100** to retain the child within the seat **100**. In particular, the restraint straps **152** can retain the child within the recess **210** of the seat support **200**. The restraint assembly **150** can also include one or more buckles **154** for fastening and/or tightening the restraint straps **152** about the child.

With reference again to FIG. **5**, the restraint straps **152** can be inserted through the restraint holes **214** defined by the seat support **200**. When the seating surface **102** is attached to the seat rim **101**, the restraint straps **152** can also be inserted through holes (not labelled) in the seating surface **102**. When a child is positioned on the seat support **200**, the restraint straps **152** can be fastened around the child to retain the child within the seat **100**. The connection of the restraint straps **152** to the seat support **200** can be a fixed connection, such that the connection between the restraint straps **152** and the seat support **200** is substantially prevented from moving due to movement of the child positioned on the seat **100**. Alternatively, the connection between the restraint straps **152** and the seat support **200** can allow for movement of the restraint straps **152** within the restraint holes **214**. The movement of the restraint straps **152** within the restraint holes **214** can allow for movement of the child within the seat **100** while still retaining the child on the seat support **200**.

The restraint holes **214** can have a slot or elongate shape. In this regard, the restraint holes **214** can have a shape that corresponds to the shape of the restraint straps **152**. The shape of the restraint holes **214** can facilitate insertion and adjustment of the restraint straps **152** with respect to the seat support **200**. FIG. **16** illustrates a bottom perspective view of a portion of the child swing **1** including the restraint straps **152** extending through the restraint holes **214**; FIG. **17** illustrates a top perspective view of a portion of the child swing **1** including the restraint straps **152**; FIG. **18** illustrates a bottom perspective view of a portion of the child swing **1** including the restraint straps **152** extending through the restraint holes **214**. The restaint straps **152** are fixed to the seat support **200** beneath the soft goods via, e.g., butterfly stitches.

FIGS. **9-15** illustrate a child swing **2**, according to another aspect of this disclosure. Portions of the alternate aspect of the child swing **2** disclosed in FIGS. **9-15** are similar to aspects of the child swing **1** described above in FIGS. **1-8** and those portions function similarly to those described above. The child swing **2** comprises a seat **300**. FIGS. **9-15** illustrate the child swing **2** without the seating surface **102** coupled to the seat **300**. The seat **300** is configured to support a child on the child swing **2**. FIG. **9** illustrates a front perspective view of the child swing **2**; FIG. **10** illustrates a side view of the child swing **2**; FIG. **11** illustrates a top perspective view of a portion of the child swing **2** including a seat support **400**; FIG. **12** illustrates a top side view of the child swing **2**. Portions of the child swing **1** described in FIGS. **1-8** including, e.g., the base **10**; the column **20**; the recline mechanism **40**; and the legs **80***a*, **80***b*, may be included in the child swing **2** described in FIGS. **9-18** in a manner substantially similar to that of the child swing **1**.

The seat **300** comprises a seat rim **301**, a support member **314**, and a seat support **400**. The seat support **400** is attached to the seat rim **301**. As shown in FIG. **12**, the seat support **400** is smaller than the seat support **200** of the child swing **1**, as shown in FIG. **4**, thereby reducing the amount of material used to manufacture the seat support **400** relative to the seat support **200**. The seat support **400** is shaped to accommodate a rear of the child. It will be appreciated that the exemplary embodiments are not limited to the shape and size described for seat supports **200** and **400** and other shapes or sizes of seat supports can be used to accommodate and support a child positioned within respective seats **100** and **300**.

FIG. **13** illustrates a bottom perspective view of a portion of the child swing **2**; FIG. **14**A illustrates an exploded view of the seat **300** of the child swing **2**; FIG. **15** illustrates a side assembled view of the seat **300** of the child swing **2**. A contact portion **430** between the seat rim **301** and the seat support **400**, as shown in FIG. **13**, includes a break **432**, as shown in FIG. **14**A. The break **232** can be defined by a location along an edge of the seat support **400** that is not in contact with the seat rim **301** and is adjacent to the contact portion **430** between the seat support **400** and the seat rim **301** along the length of the seat rim **301**. In the example shown in, e.g., FIG. **14**A, the break **432** is located at the lower end of the seat **300**.

The seat rim **301** can define a receiving channel **309** that extends around an inner perimeter of the seat rim **301**, as shown in FIG. **14**A. The receiving channel **309** can be configured to receive an outer edge of the soft goods such that the seat rim **301** remains exposed (i.e., not covered by the soft goods) when the soft goods is attached to the seat rim **301**. The soft goods can be attached to the seat rim **301** by inserting a retention member (not shown) into the receiving channel **309** to prevent the portion of the outer edge of the soft goods positioned within the receiving channel **309** from being removed (e.g., from being pulled out of the receiving channel **309**).

FIG. **14**B and **14**C illustrate perspective views of an actuator connector **302** of the seat **300** according to an aspect of this disclosure. The seat **300** includes an actuator connector **302** that can be connected to the seat rim **301** to form a perimeter about the seat **300**, as shown in FIG. **14**A. The actuator connector **302** can be connected to the seat rim **301** via bolts, screws, snap-fit, or other fastening mechanisms.

The actuator connector **302** can comprise an actuator configured to, for example, incline/recline the seat **300** relative to a base **30** of the child swing **2**. The actuator connector **302** defines an actuator channel **304** that extends around an inner perimeter of the actuator connector **302**. When the actuator connector **302** is connected to the seat rim **301**, the actuator channel **304** and the receiving channel **309** align to form a continuous channel extending around the inner perimeter of the seat rim **301** and actuator connector **302**.

The actuator channel **304** includes a channel opening **306** that opens to an interior of the seat **300**. The channel opening **306** includes a first channel opening that has a first width w_1 and a second channel opening that has a second width w_2. In an aspect, the second width w_2 is greater than the first width w_1. The difference in width of the first width w**1** and the second width w_2 can facilitate the insertion of the retention member into the receiving channel **309** of the seat rim **301** and the actuator channel **304** of the actuator connector **302**. For example, a retention member having a cross sectional dimension that is greater than the first width w_1 and less than the second width w_2 can be inserted into the actuator channel **304** through the portion of the actuator channel **304** having the second width w_2. The retention member can then be inserted into the receiving channel **309** of the seat rim **301** from the actuator channel **304**. In an aspect, the retention member can be a flexible member, such that the retention member can conform to the shape of the receiving channel **309** after insertion.

FIGS. **19-27** illustrate a child swing **3**, according to another aspect of this disclosure. Portions of the alternate aspect of the child swing **3** disclosed in FIGS. **19-27** are similar to aspects of the child swings **1** and **2** described above in FIGS. **1-18** and those portions function similarly to those described above. The child swing **3** comprises a seat **500**. FIGS. **19-24** illustrate the child swing **3** without a seating surface **502** coupled to the seat **500**. The seat **500** is configured to support a child on the child swing **3**. FIG. **19** illustrates a front perspective view of the child swing **3**; FIG. **20** illustrates a side view of the child swing **3**; FIG. **21** illustrates a top side view of the child swing **3**; FIG. **22** illustrates a top perspective view of a portion of the child swing **3**; FIG. **23**B illustrates an actuator connector **504** of the seat **500** of the child swing **3**. Portions of the child swing **1** described in FIGS. **1-8** including, e.g., the base **10**; the column **20**; the recline mechanism **40**; the seating surface **102** (e.g., soft goods); and the legs **80***a*, **80***b*, may be included in the child swing **3** described in FIGS. **19-27** in a manner substantially similar to that of the child swing **1**.

The seat **500** comprises a seat rim **501** and a seat support **600**. The seat support **600** is attached to the seat rim **501**. The seat support **600** can be attached to the seat rim **501** toward both an upper end **510** of the seat rim **501** and a lower end **512** of the seat rim **501**, such that the seat support **600** extends beneath an opening **503** defined by the seat rim **501**. The seat **500** further includes an actuator connector **504** that can be similar to the actuator connector **302** described above with regard to the seat **300** of FIGS. **9-15**. The seat support **600** is positioned and shaped to accommodate a rear of the child and also provide back support for the child. It will be appreciated that the seat support **600** can include other shapes or sizes to accommodate and support a child positioned within seat **500**.

FIG. **23**A illustrates an exploded view of a seat **500** of the child swing **3**; FIG. **24** illustrates a bottom perspective view of the seat **500** of the child swing **3**. As shown in FIG. **24**A, a contact portion **630** between the seat rim **501** and the seat support **600**, including contact portions **630***a*, **630***b* located toward the upper end **510** and contact portions **630***c*, **630***d* at the lower end **512** (partially occluded in FIG. **24** and referenced in FIG. **23**A), includes break **632***a* (located adjacent to the upper end **510** of the seat **500** between contact portions **630***a* and **630***b*), break **632***b* (located adjacent to one side (e.g., a first side) of the seat **500** between contact portions **630***b* and **630***c*), break **632***c* (located adjacent to the other side (e.g., a second side) of the seat **500** between contact portions **630***a* and **630***d*) and break **632***d* (located adjacent to the lower end **512** of the seat **500** between contact portions **630***c* and **630***d*) (occluded in FIG. **24** and referenced in FIG. **23**A). The breaks **632***a*, **632***b*, **632***c*, and **632***d* are defined by locations along an edge of the seat support **600** that is not in contact with the seat rim **501** and are adjacent to contact portions **630***a* and **630***b* between the seat support **600** and the seat rim **501** along the length of the seat rim **501**.

Adjacent to the breaks **632**, the seat support **600** can be shaped so that gaps **634** are formed between the seat rim **501** and the seat support **600**. The breaks **632***a*, **632***b*, and **632***c* and associated gaps **634***a*, **634***b* and **634***c* can be positioned between the seat rim **501** and the seat support **600** at locations to provide, for example, easier gripping and carrying the child seat **3**, space for features to be positioned along the seat rim **501** or along the edge of the seat support **600** (e.g., an actuator or other features as shown in FIG. **25**), or for other reasons. It will be appreciated that the contact portion **630** can extend between the seat rim **501** and the seat support **600** about a substantial entirety of the seat rim **501** such that there are no breaks in contact between the seat rim **501** and the seat support **600** about the edge of the seat support **600**. Alternatively, as shown above, there can be one or more breaks about between the seat rim **501** and the seat support **600**.

FIGS. **25-27** show the seat support **600** of the child swing **3** including restraint holes **614** for restraint straps **552** extending through the seat support **600**. FIG. **25** illustrates a bottom perspective view of a portion of the child swing **3**; FIG. **26** illustrates a bottom perspective view of a portion of the child swing **3**; FIG. **27** illustrates a side view of a portion of the child swing **3**. The seat support **600** can include at least one restraint hole **614** (e.g., an anchor point) that extends through the seat support **600**, as shown in FIG. **21**. The restraint holes **614** can be configured to receive, for example, the restraint straps **552** therethrough. When a seating surface **502** (e.g., soft goods for seating as described above with regard to the seating surface **102** of FIG. **1**) is attached to the seat rim **501**, the restraint straps **552** can also be inserted through holes (not labelled) in the seating surface **502**.

When a child is positioned on the seat support **600**, the restraint straps **552** can be fastened around the child to retain the child within the seat **500**. A restraint assembly may comprise a harness comprising multiple restraint straps **552** anchored to the seat support **600** via multiple restraint holes **614**, as will be described in greater detail below with regard to FIGS. **28-32**. It will be appreciated that use of the restraint holes **614** to receive the restraint straps **552** can be optional. For example, the restraint straps **552** can be inserted through holes (not labelled in FIGS. **25-27**) in the seating surface **502** and not through the restraint holes **614** to retain the child within the child swing **3**. In an aspect, the seat support **600** does not include restraint holes **614**, and the child is retained within the child swing via the restraint straps **552** inserted through the seating surface **502**.

As shown in FIGS. **26-27**, the restraint holes **614** can also receive webbing straps **554** within. The webbing straps **554** can secure the seating surface **502** to the seat support **600**. The webbing straps **554** can be attached to the seat support **600** by, e.g., a double-butterfly stitch. In some examples, one or more of the restraint holes **614** can receive both a restraint strap **552** and a webbing strap **554** within.

FIGS. **28-32** illustrate a further aspect of a child swing comprising a harness assembly **550**, according to aspects of this disclosure. The harness assembly **550** is described with regard to the child swing **3** of FIGS. **19-27**. However, it should be understood that the harness assembly **550** can be implemented in other types of child swing, including the child swing **1** and the child swing **2** described above, and the present embodiments are not limited to the child swings **1**, **2** or **3**. Additionally, the child swing of FIGS. **28-32** can be transitioned between a first mode, e.g., a mode suitable for holding an infant, and a second mode, e.g., a mode suitable for holding a child larger than an infant. Similar to the harness assembly **550**, the first and second modes are described with regard to the child swing **3** of FIGS. **19-27**. However, it should be understood that the functionality described for the first and second modes can be implemented in any of the child swings **1**, **2** or **3** or other types of child swings.

FIG. **28**A illustrates the child swing **3** in a first mode (e.g., an infant mode) with a harness assembly **550** (e.g., a restraint assembly) and infant padding **506**. The infant padding **506** can comprise soft goods with cushioning for receiving a child, e.g., an infant child. The infant padding **506** is sized and shaped for supporting children within certain size ranges, as would be understood by those skilled in the art. The infant padding **506** can be positioned on top of the seating surface **502** (e.g., soft goods) connected to the seat rim **501**. The seating surface **502** is positioned on top of the seat support **600**.

FIGS. **29**A and **29**B illustrate a connection between the harness assembly **550** and the seat support **600**, according to an aspect of this disclosure. The harness assembly **550** comprises a number of restraint straps **552**, e.g., five restraint straps **552**, attached to the seat support **600** at different locations around the seat support **600**. In this aspect, the restraint straps **552** are attached to the seat support **600** with a double-butterfly stitch to substantially prevent the restraint straps **552** from pulling through the seat support **600** and from dangling below the seat **500**. It will be appreciated that other attachment methods and/or structures can be used to attach the harness assembly **550** to the seat **500**.

The seating surface **502** comprises restraint holes **508** for the receiving the restraint straps **552** extending from the seat support **600**, as shown in FIG. **28**B. In this example, the infant padding **506** also includes restraint holes (not labeled). The loose ends of the restraint straps **552**, as shown in FIG. **28**A, extending from the seat support **600** through the seating surface **502** and the infant padding **506** can be connected (e.g., the harness assembly **500** is buckled together). In the first mode, the harness assembly **500** can be buckled together around an infant child placed in the seat **500**.

FIG. **28**B illustrates the child swing **3** in a second mode (e.g., a child mode), with the harness assembly **550** below the seating surface **602** and with the infant padding **506** removed. The harness assembly **550** can remain attached to the seat support **600** in both the first and second modes. To remove the infant padding **506**, the restraint straps **552** can be pulled through the restraint holes of the infant padding **506** and the infant padding **506** can be lifted off the seating surface **502**.

FIG. **30** illustrates the removal of the harness assembly **550** from the seating surface **502** that is attached to the seat rim **501**. Each of the restraint strap **552** of the harness assembly **550** can be pulled through the restraint holes **508** in the seating surface **502**. FIG. **31** illustrates the harness assembly **550** stored underneath the seating surface **502** on top of the seat support **600**. The harness assembly **550** can be buckled together underneath the soft goods to prevent dangling. The soft goods can be pressed down onto the seat **500** to cover the harness assembly **550**.

FIG. **32** illustrates a board **560** (e.g., polypropylene (PP) board) positioned within the seating surface **602**. The board **560** can be sewn into the soft goods to cover the harness assembly **550** to create a smooth surface for supporting a child. It will be appreciated that the board **560** can be positioned above or below the soft goods to create a smooth surface for supporting the child.

Unless explicitly stated otherwise, each numerical value and range should be interpreted as being approximate as if the word "about," "approximately," or "substantially" preceded the value or range. The terms "about," "approximately," and "substantially" can be understood as describing a range that is within 15 percent of a specified value unless otherwise stated.

Conditional language used herein, such as, among others, "can", "could", "might", "may", "e.g.," and the like, unless specifically stated otherwise, or otherwise understood within the context as used, is generally intended to convey that certain embodiments include, while other embodiments do not include, certain features, elements, and/or steps. Thus, such conditional language is not generally intended to imply that features, elements, and/or steps are in any way required for one or more embodiments or that one or more embodiments necessarily include logic for deciding, with or without author input or prompting, whether these features, elements and/or steps are included or are to be performed in any particular embodiment. The terms "comprising", "including", "having," and the like are synonymous and are used inclusively, in an open-ended fashion, and do not exclude additional elements, features, acts, operations, and so forth.

While certain example embodiments have been described, these embodiments have been presented by way of example only and are not intended to limit the scope of the inventions disclosed herein. Thus, nothing in the foregoing description is intended to imply that any particular feature, characteristic, step, module, or block is necessary or indispensable. Indeed, the novel methods and systems described herein may be embodied in a variety of other forms; furthermore, various omissions, substitutions, and changes in the form of the methods and systems described herein may be made without departing from the spirit of the inventions disclosed herein. The accompanying claims and their equivalents are intended to cover such forms or modifications as would fall within the scope and spirit of certain of the inventions disclosed herein.

The indefinite articles "a" and "an," as used herein in the specification and in the claims, unless clearly indicated to the contrary, should be understood to mean "at least one." Thus, it will be understood that reference herein to "a," "and," or "one" to describe a feature such as a component or step does not foreclose additional features or multiples of the feature. For instance, reference to a device having, comprising, including, or defining "one" of a feature does not preclude the device from having, comprising, including, or defining more than one of the feature, as long as the device has, comprises, includes, or defines at least one of the feature. Similarly, reference herein to "one of" a plurality of features does not foreclose the invention from including two or more of the features. For instance, reference to a device having, comprising, including, or defining "one of a protrusion and a recess" does not foreclose the device from having both the protrusion and the recess.

The phrase "and/or," as used herein in the specification and in the claims, should be understood to mean "either or both" of the components so conjoined, i.e., components that are conjunctively present in some cases and disjunctively present in other cases. Multiple components listed with "and/or" should be construed in the same fashion, i.e., "one or more" of the components so conjoined. Other components may optionally be present other than the components specifically identified by the "and/or" clause, whether related or unrelated to those components specifically identified. Thus, as a non-limiting example, a reference to "A and/or B", when used in conjunction with open-ended language such as "comprising" can refer, in one embodiment, to A only (optionally including components other than B); in another embodiment, to B only (optionally including components other than A); in yet another embodiment, to both A and B (optionally including other components); etc.

As used herein in the specification and in the claims, "or" should be understood to have the same meaning as "and/or" as defined above. For example, when separating items in a list, "or" or "and/or" shall be interpreted as being inclusive, i.e., the inclusion of at least one, but also including more than one, of a number or list of components, and, optionally, additional unlisted items. Only terms clearly indicated to the contrary, such as "only one of" or "exactly one of," or, when used in the claims, "consisting of," will refer to the inclusion of exactly one component of a number or list of components. In general, the term "or" as used herein shall only be interpreted as indicating exclusive alternatives (i.e. "one or the other but not both") when preceded by terms of exclusivity, such as "either," "one of," "only one of," or "exactly one of" "Consisting essentially of," when used in the claims, shall have its ordinary meaning as used in the field of patent law.

As used herein in the specification and in the claims, the phrase "at least one," in reference to a list of one or more components, should be understood to mean at least one component selected from any one or more of the components in the list of components, but not necessarily including at least one of each and every component specifically listed within the list of components and not excluding any combinations of components in the list of components. This definition also allows that components may optionally be present other than the components specifically identified within the list of components to which the phrase "at least one" refers, whether related or unrelated to those components specifically identified. Thus, as a non-limiting example, "at least one of A and B" (or, equivalently, "at least one of A or B," or, equivalently "at least one of A and/or B") can refer, in one embodiment, to at least one, optionally including more than one, A, with no B present (and optionally including components other than B); in another embodiment, to at least one, optionally including more than one, B, with no A present (and optionally including components other than A); in yet another embodiment, to at least one, optionally including more than one, A, and at least one, optionally including more than one, B (and optionally including other components); etc.

The words "inward," "outward," "upper," and "lower" refer to directions toward or away from, respectively, the geometric center of the component.

What is claimed:

1. A seat for a child swing, the child swing including a base and a column extending from the base, the base being positionable on a floor, the seat comprising:

a seat rim connectable to the column to support the seat above the floor so that the seat can move relative to the base in a swinging motion, the seat rim defining a seat opening; and

a seat support comprising a rigid material and being attached to the seat rim such that movement between the seat support and the seat rim is substantially prevented, the seat support defining a recess on one side of the seat opening,

wherein the seat support is positioned relative to the seat rim such that when a child is introduced through the seat opening from an opposing side of the seat opening, the seat support supports the child within the recess, wherein the seat opening extends along a seat rim plane, wherein when the seat is viewed in a direction substantially perpendicular to the seat rim plane, an area of the seat support is less than 50% of an area of the seat opening.

2. The seat of claim **1**, wherein the seat rim comprises a first material that extends about the seat opening, wherein the first material and the rigid material are different.

3. The seat of claim **2**, wherein the first material comprises a metal.

4. The seat of claim **2**, wherein the rigid material comprises a plastic.

5. The seat of claim **1**, wherein the seat support is attached to the seat rim via at least one fastener.

6. The seat of claim **1**, wherein the seat support contacts the seat rim along a portion of a length of the seat rim.

7. The seat of claim **6**, wherein the portion of the length of the seat support contacting the seat rim is less than half of the length of the seat rim.

8. The seat of claim **1**, wherein the seat support contacts the seat rim at multiple contact locations and forms breaks between the contact locations.

9. The seat of claim **8**, further comprising:

an actuator connected to the seat rim and located in a break between the contact locations with the seat support, wherein the actuator and the seat rim form a perimeter of the seat,

wherein the actuator is configured to adjust an incline of the seat relative to the base.

10. The seat of claim **1**, wherein the area of the seat support is less than 30% of the area of the seat opening.

11. The seat of claim **1**, wherein the seat support defines at least one restraint hole therethrough that connects the recess to an exterior of the seat.

12. The seat of claim **1**, further comprising:

a seating surface supported by the seat rim within the recess of the seat support, the seating surface comprising soft goods.

13. The seat of claim **12**, further comprising:

a child restraint assembly attached to the seat support, wherein the child restraint assembly includes at least one restraint strap positioned through at least one restraint hole to attach the at least one restraint strap to the seat support and to retain the child within the seat when a child is supported on the seat support.

14. The seat of claim **13**, wherein the seat is transitionable between a first mode suitable for an infant and a second mode suitable for a child larger than the infant.

15. The seat of claim **14**, wherein the first mode comprises the at least one restraint strap extending through the seating surface via at least one restraint hole so that the child restraint assembly can retain the child within the seat and the second mode comprises the at least one restraint strap withdrawn through the seating surface and positioned beneath the seating surface.

16. The seat of claim **15**, wherein transitioning the seat from the first mode to the second mode includes lifting the seating surface, moving the child restraint assembly between the seat support and the seating surface, buckling the child restraint assembly, and lowering the seating surface.

17. The seat of claim **15**, wherein the seating surface comprises a polypropylene board positioned within or adjacent to the seating surface for covering the child restraint assembly when the seat is in the second mode.

18. The seat of claim **14**, wherein, in the first mode, at least a portion of the soft goods comprises infant padding positioned above the seating surface, the infant padding comprising at least one further restraint hole configured to receive the at least one restraint strap.

19. The seat of claim **1**, wherein the seat support defines at least one vent hole therethrough that connects the recess to an exterior of the seat.

20. A seat for a child swing, the child swing including a base and a column extending from the base, the base being positionable on a floor, the seat comprising:

a seat rim connectable to the column to support the seat above the floor, the seat rim defining a seat opening;

a seat support comprising a rigid material and being attached to the seat rim such that movement between the seat support and the seat rim is substantially prevented, the seat support defining a recess on one side of the seat opening, wherein the seat support is positioned relative to the seat rim such that when a child is introduced through the seat opening from an opposing side of the seat opening, the seat support supports the child within the recess, wherein the seat support contacts the seat rim at multiple contact locations and forms breaks between the contact locations; and

an actuator connected to the seat rim and located in a break between the contact locations with the seat support, wherein the actuator and the seat rim form a perimeter of the seat, and wherein the actuator is configured to adjust an incline of the seat relative to the base.

21. A seat for a child swing, the child swing including a base and a column extending from the base, the base being positionable on a floor, the seat comprising:

a seat rim connectable to the column to support the seat above the floor, the seat rim defining a seat opening;

a seat support comprising a rigid material and being attached to the seat rim such that movement between the seat support and the seat rim is substantially prevented, the seat support defining a recess on one side of the seat opening, wherein the seat support is positioned relative to the seat rim such that when a child is introduced through the seat opening from an opposing side of the seat opening, the seat support supports the child within the recess;

a seating surface supported by the seat rim within the recess of the seat support, the seating surface comprising soft goods; and

a child restraint assembly attached to the seat support, wherein the child restraint assembly includes at least one restraint strap positioned through at least one restraint hole to attach the at least one restraint strap to the seat support and to retain the child within the seat when a child is supported on the seat support, wherein the seat is transitionable between a first mode suitable for an infant and a second mode suitable for a child larger than the infant and wherein the first mode comprises the at least one restraint strap extending through the seating surface via at least one restraint hole so that the child restraint assembly can retain the child within the seat and the second mode comprises the at least one restraint strap withdrawn through the seating surface and positioned beneath the seating surface.

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